

Oscillations In The Kaon Sector And Neutrino Sector: QISKIT Implementation

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submitted by

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Abstract

This report explores the problem of 2-body and 3-body classical coupled oscillator systems, the framework of which will be used in the study of neutrino oscillations. The coupled oscillator problem has been solved in detail. This report also goes through important concepts of linear algebra and vector spaces. Using the formalism, the time evolution of coupled oscillators is solved through eigenvalue problem. The report then transitions to a quantum field theory perspective, exploring flavour oscillations and CP violation in the quark and the lepton sector. A contrast has been provided between the neutral kaon system and the neutrino sector. The PMNS matrix is introduced in the 3 flavour neutrino oscillations, whose properties are discussed that give rise to CP violation. A brief introduction to quantum computing is provided. After this the report simulated kaon decays using a quantum circuit simulation by using QISKIT module. Then the 2 flavour neutrino oscillations have also been simulated using QISKIT. The results and findings from the simulation and the theory have been finally summarized. The future aspects of the report also have been discussed in the summary.

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Chapter 1

Preliminaries

To build a strong foundation in quantum physics, the study began with linear algebra, which serves as a vital mathematical framework for various concepts and operations in quantum mechanics. Linear algebra enables the representation of quantum states and observables, describes their time evolution and formulates measurement processes. Its tools and structures are essential for understanding and working with quantum systems, making it a fundamental aspect of quantum mechanics.

1.1 Fields & Vector spaces

1.1.1 Fields

A field ($\mathbb{F}$) is a set with the following two operations defined $\forall a, b, c \in \mathbb{F}$:

1. **Addition** $(+): \mathbb{F} \times \mathbb{F} \rightarrow \mathbb{F}$
2. **Multiplication** $(\cdot, \times): \mathbb{F} \times \mathbb{F} \rightarrow \mathbb{F}$

such that the following field axioms and the distributive property hold:

1. Field Axioms For Addition:

- **Closure** : $a + b \in \mathbb{F}$
- **Associativity** : $(a + b) + c = a + (b + c)$
- **Commutativity** : $a + b = b + a$
- **Identity element** : $\exists 0 \in \mathbb{F}$ such that $a + 0 = a$
- **Additive Inverse** : $\exists -a \in \mathbb{F}$ such that $a + (-a) = 0$

2. Field Axioms For Multiplication (Excluding 0):

- **Closure** : $a \cdot b \in \mathbb{F}$
- **Associativity** : $(a \cdot b) \cdot c = a \cdot (b \cdot c)$
- **Commutativity** : $a \cdot b = b \cdot a$
- **Identity element** : $\exists 1 \in \mathbb{F}$ such that $a \cdot 1 = a$

- **Multiplicative Inverse** : $\exists a^{-1} \in \mathbb{F}$ such that $a \cdot (a^{-1}) = 1$

3. Distributive Property:

- $a \cdot (b + c) = a \cdot b + a \cdot c$

Examples of fields: $\mathbb{R}$ (real numbers), $\mathbb{Q}$ (rational numbers), $\mathbb{C}$ (complex numbers), $\mathbb{Z}_p$ (integers modulo a prime p), etc.

Vectors

In the context of our study, from now on we will define vectors as the elements of the vector space. They are abstract mathematical objects that follow certain rules defined by the structure of the space.

1.1.2 Vector Space

A vector space V is defined over a field $\mathbb{F}$ (which for most of our cases would be $\mathbb{R}$ or $\mathbb{C}$) as a set on which two operations are defined: addition of vectors and multiplication by scalars, such that for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and $a, b \in \mathbb{F}$, the following axioms hold [1]:

1. $\mathbf{u} + \mathbf{v} \in V$ (Closure under addition)
2. $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$ (Commutativity of addition)
3. $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$ (Associativity of addition)
4. $\exists \mathbf{0} \in V$, such that $\mathbf{u} + \mathbf{0} = \mathbf{u}$ (Additive identity exists)
5. $\forall \mathbf{u} \in V, \exists -\mathbf{u} \in V$, such that $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$ (Additive inverse exists)
6. $a\mathbf{u} \in V$ (Closure under multiplication)
7. $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$ (Distributivity of scalar multiplication over vector addition)
8. $(a + b)\mathbf{u} = a\mathbf{u} + b\mathbf{u}$ (Distributivity of scalar addition over scalar multiplication)
9. $(ab)\mathbf{u} = a(b\mathbf{u})$ (Associativity of scalar multiplication)
10. $1\mathbf{u} = \mathbf{u}$ (Multiplicative identity of scalar multiplication)

Examples of vector space:

- $\mathbb{R}^n$: the set of all n -tuples of real numbers.
- The set of all polynomials with real coefficients
- The set of all continuous real-valued functions on an interval.

If the field over which the vector space is defined are Real numbers, i.e $\mathbb{F} = \mathbb{R}$ then V is called a real vector space. If i.e $\mathbb{F} = \mathbb{C}$ then V is called a complex vector space.

Linear Independence

Let V be a vector space over the scalars $\mathbb{F}$. A set of unique vectors $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\} \in V$ are said to be linearly independent if the equation ($a_1, a_2, \dots, a_n \in \mathbb{F}$):

$$a_1 \mathbf{u}_1 + a_2 \mathbf{u}_2 + \dots + a_n \mathbf{u}_n = 0 \quad (1.1)$$

has solution only for $a_1 = a_2 = \dots = a_n = 0$. This means if we take any vector from the set of linearly independent vectors, then it can not be expressed as a linear combination of other vectors from that set.

Dimension of a vector space

The dimension of a vector space n is the at most number of linearly independent vectors it can have. It also depends on the choice of the scalar field over which the space is defined.

Basis

A *basis* is a maximal linearly independent set of vectors that span the entire space. That is, if $\beta = \{v_1, v_2, \dots, v_k\}$ is a basis of V then any vector $\mathbf{v} \in V$ has a unique decomposition as the linear combination of vectors in β .

$$\mathbf{v} = c_1 v_1 + c_2 v_2 + \dots + c_k v_k, \quad c_i \in \mathbb{F}$$

The choice of basis is not necessarily unique, for different choice of basis vectors the coefficients c_i would be different.

A n -dimensional vector space would have exactly n basis vectors.

Subspace

A subset $W \subset V$ which is a vector space in itself over the same field of scalars is called a subspace.

Dual space

The dual space of a vector space V is denoted as V^* , is the set of all linear functionals from V to $\mathbb{F}$.

$$V^* = \{f : V \rightarrow \mathbb{F} | f \text{ is linear}\}$$

That is, $f \in V^*$ satisfies:

$$f(\alpha v + \beta w) = \alpha f(v) + \beta f(w) \quad \text{for all } v, w \in V, \alpha, \beta \in \mathbb{F}$$

Vector spaces inherently have no notion of length or direction defined for the elements in them. But we can define a quantity which can be used to define a length between the elements of a vector space. Such a quantity is called the *norm*.

1.2 Dirac Notation

In quantum mechanics and Hilbert space theory, vectors and linear functionals are commonly represented using **Dirac notation**. The basic elements are:

- A **ket** $|\psi\rangle$ represents a vector ψ in a vector space.
- A **bra** $\langle\phi|$ represents a linear functional (element of the dual space) acting on vectors.
- The inner product (a complex number) of two vectors ϕ and ψ is written as:

$$\langle\phi|\psi\rangle$$

- The outer product $|\psi\rangle\langle\phi|$ represents a linear operator.

For example, if our vector space is: $\mathbb{R}^n$ then the ket vectors $|v\rangle$ represent a $n \times 1$ matrix (column vector $\in V$) while the bra vectors $\langle v|$ represent a $1 \times n$ matrix (row vector $\in V^*$)

Ket:	$ \psi\rangle$
Bra:	$\langle\phi $
Inner product:	$\langle\phi \psi\rangle$
Outer product:	$ \psi\rangle\langle\phi $
Operator acting on a ket:	$\hat{A} \psi\rangle$
Expectation value:	$\langle\psi \hat{A} \psi\rangle$

The dirac notations would be used for convenience from here on.

1.3 Inner product spaces and Hilbert space

1.3.1 Inner Product Spaces

An **inner product space** is a vector space V over the field $\mathbb{F}$ (where $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$) equipped with an additional structure called an *inner product*. An inner product is a function

$$\langle\cdot, \cdot\rangle : V \times V \rightarrow \mathbb{F}$$

that satisfies the following properties for all $u, v, w \in V$ and $\alpha \in \mathbb{F}$:

1. **Conjugate Symmetry:** $\langle u, v \rangle = \overline{\langle v, u \rangle}$
2. **Linearity in the first argument:** $\langle \alpha u + v, w \rangle = \alpha \langle u, w \rangle + \langle v, w \rangle$
3. **Positive semi-definiteness:** $\langle v, v \rangle \geq 0$, and equality holds if and only if $v = 0$

The inner product allows us to define concepts such as the **length (norm)** of a vector:

$$\|v\| = \sqrt{\langle v, v \rangle}$$

and the **angle** between vectors, as well as the notion of **orthogonality** ($\langle u, v \rangle = 0$).

1.3.2 Hilbert Spaces

A **Hilbert space** is a complete inner product space. This means it is a vector space $\mathcal{H}$ equipped with an inner product $\langle \cdot, \cdot \rangle$, and it is **complete** with respect to the norm induced by this inner product. Completeness means that every Cauchy sequence in $\mathcal{H}$ converges to a limit that also lies in $\mathcal{H}$.

Hilbert spaces generalize the familiar Euclidean spaces $\mathbb{R}^n$ to possibly infinite-dimensional settings, and they form the mathematical foundation for many areas of functional analysis and quantum mechanics.

Examples of Hilbert spaces include:

- $\mathbb{R}^n$ and $\mathbb{C}^n$ with the standard dot product.
- The space $L^2([a, b])$ of square-integrable functions on an interval.

1.4 Operators

In linear algebra and quantum mechanics, an **operator** is a rule or function that acts on elements of a vector space and returns another element of the same (or a related) vector space [2].

1.4.1 Linear Operators

A function $\hat{A} : V \rightarrow V$ is called a **linear operator** if it satisfies the following two properties for all vectors $|u\rangle, |v\rangle \in V$ and scalars $\alpha \in \mathbb{F}$:

$$\hat{A}(|u\rangle + |v\rangle) = \hat{A}(|u\rangle) + \hat{A}(|v\rangle) \quad (\text{additivity})$$

$$\hat{A}(\alpha |v\rangle) = \alpha \hat{A}(|v\rangle) \quad (\text{homogeneity})$$

That is, linear operators preserve vector addition and scalar multiplication.

In many contexts, especially in physics, operators are denoted with a “hat”, such as $\hat{A}, \hat{H}, \hat{X}$, etc. It is not necessary that operators commute with other operators.

$$\hat{A}\hat{B} \neq \hat{B}\hat{A} \quad \text{not necessarily true.}$$

We define the commutator of two matrices as follows:

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$$

The matrix representation of a linear operator $\hat{X}$ with respect to a basis $\{e_i\}$ of V is given by $\hat{X}_{ij} = \langle e_i, \hat{X}e_j \rangle$.

In quantum mechanics:

- Operators act on state vectors (kets): $\hat{A}|\psi\rangle$
- The **eigenvalue equation** is given by:

$$\hat{A}|\psi\rangle = \lambda|\psi\rangle$$

where λ is an eigenvalue and $|\psi\rangle$ is the corresponding eigenvector (or eigenstate).

- The **expectation value** of an operator $\hat{A}$ in state $|\psi\rangle$ is:

$$\langle \hat{A} \rangle = \langle \psi | \hat{A} | \psi \rangle$$

Examples

- The identity operator $\hat{I}$ satisfies $\hat{I} |\psi\rangle = |\psi\rangle$
- The differential operator $\hat{D} = \frac{d}{dx}$
- The Hamiltonian $\hat{H}$ governs the time evolution in quantum mechanics

1.4.2 Hermitian, anti-Hermitian & Unitary operators

These are special cases of operators which satisfy certain properties. It is in our interest to go over few such cases.

Hermitian

An operator $\hat{A}$ on a Hilbert space $\mathcal{H}$ is *Hermitian* (or self-adjoint) if:

$$\langle \phi | \hat{A} \psi \rangle = \langle \hat{A} \phi | \psi \rangle \text{ for all } |\psi\rangle, |\phi\rangle \in \mathcal{H}$$

In matrix form:

$$\hat{A}^\dagger = \hat{A}$$

where $\hat{A}^\dagger$ is the adjoint or conjugate transpose of $\hat{A}$.

Properties:

- All eigenvalues of a Hermitian operator are real.
- Eigenvectors corresponding to distinct eigenvalues are orthogonal.
- Hermitian operators can be diagonalized by a complete orthonormal basis.
- The eigenvalues of hermitian operators represent observables (measurable quantities) in quantum mechanics, example: Energy $\hat{H}$ (Hamiltonian), Momentum $\hat{p}$, etc.

Anti-Hermitian

An operator $\hat{A}$ on a Hilbert space $\mathcal{H}$ is *anti-Hermitian* (or skew-Hermitian) if:

$$\langle \phi | \hat{A} \psi \rangle = -\langle \hat{A} \phi | \psi \rangle \text{ for all } |\psi\rangle, |\phi\rangle \in \mathcal{H}$$

In matrix form:

$$\hat{A}^\dagger = -\hat{A}$$

Properties:

- All eigenvalues of an anti-Hermitian operator are purely imaginary or 0.
- Anti-Hermitian operators can be acquired by multiplying a hermitian operator by i .
- Anti-Hermitian operators generate unitary time evolution in quantum mechanics- preserving probability and inner product.

Unitary

An operator $\hat{U}$ is called **unitary** if it preserves inner products, that is:

$$\hat{U}^\dagger \hat{U} = \hat{U} \hat{U}^\dagger = \hat{I}$$

This means that $\hat{U}^{-1} = \hat{U}^\dagger$, so $\hat{U}$ is invertible and its inverse is its adjoint.

Properties:

- **Preserves norms:**

$$\|\hat{U} |\psi\rangle\| = \| |\psi\rangle \|$$

- **Preserves inner products:**

$$\langle \phi | \psi \rangle = \langle \hat{U} \phi | \hat{U} \psi \rangle$$

- **Eigenvalues** of a unitary operator lie on the **unit circle** in the complex plane. That is, if $\hat{U} |\psi\rangle = \lambda |\psi\rangle$, then $|\lambda| = 1$.
- Unitary operators describe **time evolution** and **symmetries** in quantum mechanics.
- The Schrödinger equation leads to unitary time evolution:

$$|\psi(t)\rangle = \hat{U}(t) |\psi(0)\rangle$$

- In quantum computing, all quantum gates are represented by unitary operators.
- Symmetry operations such as rotations, translations, and Lorentz transformations are typically represented by unitary operators.

Chapter 2

The Coupled Oscillator Problem

When two oscillating systems are connected in a manner which allows for transfer of energy between them is called a coupled oscillator system. The study of coupled oscillators will help us in getting insights and drawing analogies to neutrino oscillations. In this chapter we will use our knowledge of vector spaces to solve the 2 body coupled oscillator and 3 body coupled oscillator problem.

2.1 Two Body coupled oscillator

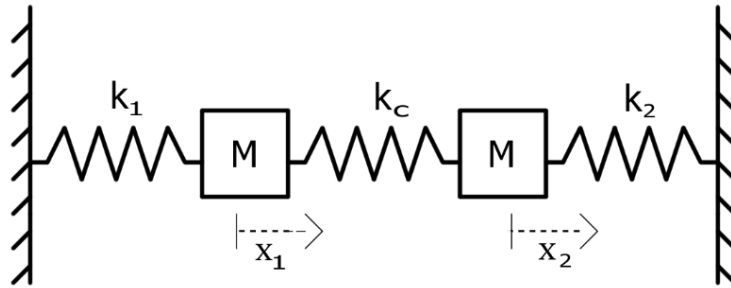


Figure 2.1: A 2 body coupled oscillator system with the spring with spring constant k_c as the coupling agent

Assume masses are at equilibrium position and then displaced x_1 & x_2 distance respectively. For simplicity we have considered the two masses to be equal. We write the equation of motion for the masses:

$$m\ddot{x}_1 = -(k_1 + k_c)x_1 + k_c x_2$$

$$m\ddot{x}_2 = k_c x_1 - (k_2 + k_c)x_2$$

We write these equations in matrix notation and turn the problem into an eigenvalue problem.

$$\mathbf{M}|\ddot{x}\rangle = -\mathbf{K}|x\rangle \quad (2.1)$$

$$\begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix} \begin{bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{bmatrix} = - \begin{bmatrix} k_1 + k_c & -k_c \\ -k_c & k_2 + k_c \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad (2.2)$$

Here $\mathbf{M}$ is the mass matrix. The general solution of the equation takes the following form:

$$|x\rangle = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} e^{-i\omega t}$$

which gives rise to:

$$|\ddot{x}\rangle = -\omega^2 \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} e^{-i\omega t}$$

Substituting this in equation (2.2) and simplifying after rearrangement we get:

$$\begin{bmatrix} k_1 + k_c - m\omega^2 & -k_c \\ -k_c & k_2 + k_c - m\omega^2 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = 0 \quad (2.3)$$

For non trivial solution for $|a\rangle = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$:

$$\begin{vmatrix} k_1 + k_c - m\omega^2 & -k_c \\ -k_c & k_2 + k_c - m\omega^2 \end{vmatrix} = 0 \quad (2.4)$$

$$\implies m^2\omega^4 - m\omega^2(k_1 + k_2 + 2k_c) + (k_1k_2 + k_1k_c + k_2k_c) = 0$$

Solving for ω^2 we get 2 roots:

$$\omega_{\pm}^2 = \frac{(k_1 + k_2 + 2k_c) \pm \sqrt{(k_1 + k_2 + 2k_c)^2 - 4(k_1k_2 + k_1k_c + k_2k_c)}}{2m} \quad (2.5)$$

Substituting $\omega_{\pm}^2$ in equation 2.3 we get two solutions for $|a\rangle$:

$$|a_{\pm}\rangle = c_{\pm} \begin{bmatrix} \frac{k_c}{k_1 + k_c - m\omega_{\pm}^2} \\ 1 \end{bmatrix}, \text{ or } |a_{\pm}\rangle = c_{\pm} \begin{bmatrix} 1 \\ \frac{k_c}{k_2 + k_c - m\omega_{\pm}^2} \end{bmatrix}, \quad c_{\pm} \text{ is a constant} \quad (2.6)$$

$\omega_{\pm}$ are the normal mode frequencies of the system. Any state of the system can be expressed as a superposition of the normal modes. A normal mode is a natural pattern of vibration where all components of the system move sinusoidally with the same frequency and constant phase difference. Writing an explicit solution:

$$|x(t)\rangle = c_+ |a_+\rangle \cos(\omega_+ t + \phi_1) + c_- |a_-\rangle \cos(\omega_- t + \phi_2) \quad (2.7)$$

The particular solution can be found by finding the constants $c_{\pm}, \phi_1, \phi_2$ by using the initial conditions for $|x(0)\rangle$ & $|\dot{x}(0)\rangle$.

2.1.1 **Case:** $k = k_1 = k_2 \neq k_c$

Finding the two different value for $\omega_{\pm}^2$:

$$\omega_+ = \sqrt{\frac{k + 2k_c}{m}} \quad \omega_- = \sqrt{\frac{k}{m}}$$

Substituting the value in equation 2.3 we get:

$$|a_+\rangle = c_+ \begin{bmatrix} -1 \\ 1 \end{bmatrix} \quad |a_-\rangle = c_- \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

If we apply the initial condition:

$$\begin{aligned}x_1(0) &= \mathbf{a}_1 & \dot{x}_1(0) &= 0 \\x_2(0) &= \mathbf{a}_2 & \dot{x}_2(0) &= 0\end{aligned}$$

Then the solution for the time evolution of the coupled oscillator system described by the conditions assumed is:

$$\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \begin{bmatrix} \frac{\mathbf{a}_1 - \mathbf{a}_2}{2} \cos\left(\sqrt{\frac{k+2k_c}{m}} t\right) + \frac{\mathbf{a}_1 + \mathbf{a}_2}{2} \cos\left(\sqrt{\frac{k}{m}} t\right) \\ \frac{\mathbf{a}_2 - \mathbf{a}_1}{2} \cos\left(\sqrt{\frac{k+2k_c}{m}} t\right) + \frac{\mathbf{a}_1 + \mathbf{a}_2}{2} \cos\left(\sqrt{\frac{k}{m}} t\right) \end{bmatrix} \quad (2.8)$$

2.1.2 Case: $k = k_1 = k_2 = k_c$

We just substitute $k_c = k$ in equation 2.8 and get the equation of motion for the same initial conditions:

$$\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \begin{bmatrix} \frac{\mathbf{a}_1 - \mathbf{a}_2}{2} \cos\left(\sqrt{\frac{3k}{m}} t\right) + \frac{\mathbf{a}_1 + \mathbf{a}_2}{2} \cos\left(\sqrt{\frac{k}{m}} t\right) \\ \frac{\mathbf{a}_2 - \mathbf{a}_1}{2} \cos\left(\sqrt{\frac{3k}{m}} t\right) + \frac{\mathbf{a}_1 + \mathbf{a}_2}{2} \cos\left(\sqrt{\frac{k}{m}} t\right) \end{bmatrix} \quad (2.9)$$

2.2 Three body coupled oscillator

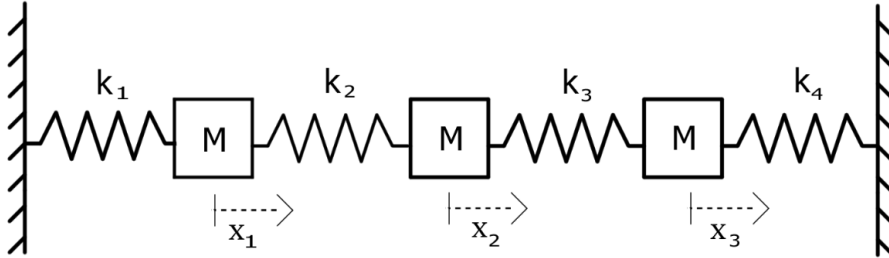


Figure 2.2: A 3 body coupled oscillator system with 4 springs

Now the sophistication has increased as we have 3 masses and 4 springs. Now we have to deal with a 3×3 matrix. The equation of motion are:

$$m\ddot{x}_1 = -(k_1 + k_2)x_1 + k_2x_2$$

$$m\ddot{x}_2 = k_2x_1 - (k_2 + k_3)x_2 + k_3x_3$$

$$m\ddot{x}_3 = k_3x_2 - (k_3 + k_4)x_3$$

In matrix notation:

$$\mathbf{M} |\ddot{x}\rangle = -\mathbf{K} |x\rangle \quad (2.10)$$

$$\Rightarrow \begin{bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \\ \ddot{x}_3 \end{bmatrix} = - \begin{bmatrix} \frac{k_1+k_2}{m} & -\frac{k_2}{m} & 0 \\ -\frac{k_2}{m} & \frac{k_2+k_3}{m} & -\frac{k_3}{m} \\ 0 & -\frac{k_3}{m} & \frac{k_3+k_4}{m} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad (2.11)$$

Again substituting $x_i = a_i e^{-i\omega t}$ for $i = 1, 2, 3$ we convert the problem to an eigenvalue problem for determining ω^2 as before.

$$\begin{bmatrix} k_1 + k_2 - m\omega^2 & -k_2 & 0 \\ -k_2 & k_2 + k_3 - m\omega^2 & -k_3 \\ 0 & -k_3 & k_3 + k_4 - m\omega^2 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = 0 \quad (2.12)$$

For non trivial solution for $|a\rangle = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$:

$$\begin{vmatrix} k_1 + k_2 - m\omega^2 & -k_2 & 0 \\ -k_2 & k_2 + k_3 - m\omega^2 & -k_3 \\ 0 & -k_3 & k_3 + k_4 - m\omega^2 \end{vmatrix} = 0 \quad (2.13)$$

Define $z = m\omega^2$ and simplify the determinant to get:

$$(k_1 + k_2 - z) [(k_2 + k_3 - z)(k_3 + k_4 - z) - k_3^2] - k_2^2(k_3 + k_4 - z) = 0$$

$$z^3 - z^2(k_1 + 2k_2 + 2k_3 + k_4) + z(k_1k_2 + 2k_1k_3 + k_1k_4 + 3k_2k_3 + 2k_2k_4 + k_3k_4) - (k_1k_2k_3 + k_1k_2k_4 + k_1k_3k_4 + k_2k_3k_4) = 0 \quad (2.14)$$

Writing the roots for z as z_1, z_2 and z_3 yields 3 values for ω^2 :

$$\omega_1^2 = z_1/m, \quad \omega_2^2 = z_2/m, \quad \omega_3^2 = z_3/m$$

The interesting thing to note here is that the roots for equation 2.14 can have 2 imaginary roots. The differential equation 2.10 for the imaginary value of ω^2 has solution of the form:

$$x_j = a_j e^{\pm kt} e^{-ibt}$$

Such a solution is not a stable solution. But luckily as $k_i > 0$ the equation 2.14 doesn't have imaginary solutions. Apparently this particular geometry of the problem under the condition $k_i > 0$ ensures there are no degenerate eigenvalues. Thus we don't have to worry about any dying down or blowing up term in our solution or repeating eigenvalues; degeneracy. Substitute the three different value of ω^2 in equation 2.12 and find the eigenvectors $|a\rangle$. Lets denote these 3 eigenvectors as $|a\rangle_x, |a\rangle_y$ & $|a\rangle_z$. The general equation of motion is a superposition of the 3 normal modes of the system. Writing down the solution:

$$|x(t)\rangle = c_x |a\rangle_x \cos(\omega_1 t + \phi_1) + c_y |a\rangle_y \cos(\omega_2 t + \phi_2) + c_z |a\rangle_z \cos(\omega_3 t + \phi_3) \quad (2.15)$$

The particular solution for the equation of motion can be found using the initial conditions ($|x(0)\rangle$ & $|\dot{x}(0)\rangle$) to get the values for the constants in the equation 2.15: $c_x, c_y, c_z, \phi_1, \phi_2$ & ϕ_3 .

2.2.1 **Case:** $k_1 = k_2 = k_3 = k_4 = k$

This is a special case where all the springs have the same spring constant= k . This simplifies equation 2.14 to:

$$(z - 2k) ((z - 2k)^2 - 2k^2) = 0$$

This gives us the three normal mode frequencies as follows:

$$w_1 = \sqrt{\frac{2k}{m}}, \quad w_2 = \sqrt{\frac{(2 + \sqrt{2})k}{m}}, \quad w_3 = \sqrt{\frac{(2 - \sqrt{2})k}{m}}$$

plugging these values in equation 2.12 we find the normal modes corresponding to each frequency:

$$w_1 \rightarrow |a\rangle_x = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ -\frac{1}{\sqrt{2}} \end{bmatrix}, \quad w_2 \rightarrow |a\rangle_y = \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{\sqrt{2}} \\ \frac{1}{2} \end{bmatrix}, \quad w_3 \rightarrow |a\rangle_z = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{\sqrt{2}} \\ \frac{1}{2} \end{bmatrix} \quad (2.16)$$

w_1 corresponds to the case in which the first and the third mass are initially displaced equal and opposite amount. Mass-2 remains stationary as it always experiences equal and opposite amount of force due to the coupling.

w_2 corresponds to the case in which mass-1 and 3 are initially displaced equally in the same direction while mass-2 in the opposite direction.

w_3 corresponds to the case in which mass-1 and 3 are initially displaced equally in the same direction while mass-2 is also displaced in the same direction.

Any other initial condition can be expressed as a superposition of these 3 normal modes. Let us assume an arbitrary displacement of the 3 degree of freedoms initially, all masses are released from rest ($\phi_{1,2,3} = 0$). After re-scaling we get:

$$|x(t)\rangle = c_x \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \cos(w_1 t) + c_y \begin{bmatrix} 1 \\ -\sqrt{2} \\ 1 \end{bmatrix} \cos(w_2 t) + c_z \begin{bmatrix} 1 \\ \sqrt{2} \\ 1 \end{bmatrix} \cos(w_3 t) \quad (2.17)$$

$$\begin{aligned} c_x + c_y + c_z &= x_1(0) \\ -\sqrt{2}c_y + \sqrt{2}c_z &= x_2(0) \\ -c_x + c_y + c_z &= x_3(0) \end{aligned}$$

Thus our equation of motion can be written as:

$$\begin{aligned} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} &= \frac{x_1(0) - x_3(0)}{2} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \cos\left(\sqrt{\frac{2k}{m}}t\right) \\ &+ \frac{x_1(0) + x_3(0) - \sqrt{2}x_2(0)}{4} \begin{bmatrix} 1 \\ -\sqrt{2} \\ 1 \end{bmatrix} \cos\left(\sqrt{\frac{(2 + \sqrt{2})k}{m}}t\right) \\ &+ \frac{x_1(0) + x_3(0) + \sqrt{2}x_2(0)}{4} \begin{bmatrix} 1 \\ \sqrt{2} \\ 1 \end{bmatrix} \cos\left(\sqrt{\frac{(2 - \sqrt{2})k}{m}}t\right) \end{aligned} \quad (2.18)$$

Note that we solved for a very specific case. 3 body coupled oscillator can be complex and have degenerate eigenvalues for w^2 in equation 2.12 arising from the geometry of the problem. We have solved for a linear geometry and positive spring constants, thus we were able to avoid any degeneracy or imaginary eigenvalues. While a problem with circular or triangular geometry might have two different modes having equal frequencies.

Chapter 3

Oscillations in Quark and Lepton sector

3.1 The neutral Kaon system

K^0 is a neutral kaon with quark composition $d\bar{s}$ belonging to the meson family. The anti neutral kaon $\bar{K}^0$ has the decomposition $\bar{d}s$. The K^0 and $\bar{K}^0$ have strangeness $S = 1$ and $S = -1$ respectively. Since strangeness is conserved for strong and electromagnetic interactions and since the kaon is the lightest strange particle, the decay of K^0 and $\bar{K}^0$ can occur only due to weak interactions.

The W boson only couples to lefthanded fermions and righthanded anti-fermions. CP turns a lefthanded fermions to righthanded antifermions, thus, weak decays are expected to be eigenstates of CP. K^0 and $\bar{K}^0$ are not eigenstates of CP, but instead are transformed into one another by CP operation. Following the convention that [3]

$$CP |K^0\rangle = |\bar{K}^0\rangle \quad CP |\bar{K}^0\rangle = |K^0\rangle \quad (3.1)$$

We can construct linear combinations of K^0 and $\bar{K}^0$, which are eigenstates of CP

$$|K_1^0\rangle = \frac{1}{\sqrt{2}}[|K^0\rangle + |\bar{K}^0\rangle] \quad CP = +1, \quad |K_2^0\rangle = \frac{1}{\sqrt{2}}[|K^0\rangle - |\bar{K}^0\rangle] \quad CP = -1 \quad (3.2)$$

Alternatively, we can write K^0 and $\bar{K}^0$ (Eigenstates of strangeness) in terms of eigenstates of CP:

$$|K^0\rangle = \frac{1}{\sqrt{2}}[|K_1^0\rangle + |K_2^0\rangle], \\ |\bar{K}^0\rangle = \frac{1}{\sqrt{2}}[|K_1^0\rangle - |K_2^0\rangle].$$

3.1.1 Decay modes

The neutral kaons ($|K^0\rangle$ & $|\bar{K}^0\rangle$) overtime evolve into the CP eigenstates $|K_1^0\rangle$ & $|K_2^0\rangle$ which over time decay into either $2\pi(\pi^+\pi^-, \pi^0\pi^0)$ or $3\pi(\pi^+\pi^-\pi^0, \pi^0\pi^0\pi^0)$. The decay products have definite CP and are the eigenstates of CP. We can associate specific lifetimes to these different decay modes. $K \rightarrow 3\pi$ has lesser phase space integral than $K \rightarrow 2\pi$. Hence, it is expected that the K_2^0 component has a longer lifetime [3]. It is also experimentally found that, short (K_s^0) and long (K_L^0) long lived components have lifetimes $\tau_s = 0.892 \times 10^{-10}\text{s}$ and $\tau_L = 5.18 \times 10^{-8}\text{s}$ respectively.

If we ignore the possibility of CP violation and study the time evolution of an initial pure beam of K^0 :

$$\text{at time } t = 0, |\psi(0)\rangle = \frac{1}{\sqrt{2}}[|K_s^0\rangle + |K_L^0\rangle]$$

The effective Hamiltonian for kaon decays is given by:

$$H_{eff} = M - i\Gamma/2$$

Thus the time evolution of the initial pure beam can be written as

$$|\psi(t)\rangle = \frac{1}{\sqrt{2}}[e^{-i\lambda_s t} |K_s^0\rangle + e^{-i\lambda_L t} |K_L^0\rangle] \quad (3.3)$$

Where $\lambda_s t$ and $\lambda_L t$ are given by:

$$\lambda_s = m_s - i\Gamma_s/2, \quad \lambda_L = m_L - i\Gamma_L/2$$

m_s & m_L are masses of K_s^0 and K_L^0 . Γ_s & Γ_L are decay widths of them. We can rewrite in the flavour basis as:

$$|\psi(t)\rangle = \frac{1}{2}[(e^{-i\lambda_s t} + e^{-i\lambda_L t}) |K^0\rangle + (e^{-i\lambda_s t} - e^{-i\lambda_L t}) |\bar{K}^0\rangle]$$

The fraction of K^0 particles in the beam, after a time t , is:

$$\begin{aligned} F(K^0) &= |\langle K^0 | \psi(t) \rangle|^2 \\ &= \frac{1}{4}[e^{-\Gamma_s t} + e^{-\Gamma_L t} + 2 \cos(\Delta m t) e^{-\Gamma t}] \end{aligned} \quad (3.4)$$

where $\Gamma = (\Gamma_s + \Gamma_L)/2$ and $\Delta m = m_L - m_s$. The corresponding fraction of $\bar{K}^0$ is:

$$\begin{aligned} F(\bar{K}^0) &= |\langle \bar{K}^0 | \psi(t) \rangle|^2 \\ &= \frac{1}{4}[e^{-\Gamma_s t} + e^{-\Gamma_L t} - 2 \cos(\Delta m t) e^{-\Gamma t}] \end{aligned} \quad (3.5)$$

Experimentally the values for the relevant parameters have been measured as follows:

- $\Gamma_s = 1.12 \times 10^{10} s^{-1}$
- $\Gamma_L = 1.93 \times 10^7 s^{-1}$
- $\Delta m = 0.535 \times 10^{10} s^{-1}$
- decay length for $K_s^0 = 5.4 \text{ cm}/(GeV/c)$
- decay length for $K_L^0 = 5.4 \text{ cm}/(GeV/c)$

The oscillatory term in equation 3.4 and 3.5 have frequency $\Delta m/2\pi$. The exponential factor multiplied to the oscillatory term leads to a pure K_L^0 beam after several K_s^0 lifetimes. The fraction of K^0 and $\bar{K}^0$ as a function of time for an initially pure beam of K^0 is given in plot 3.1.

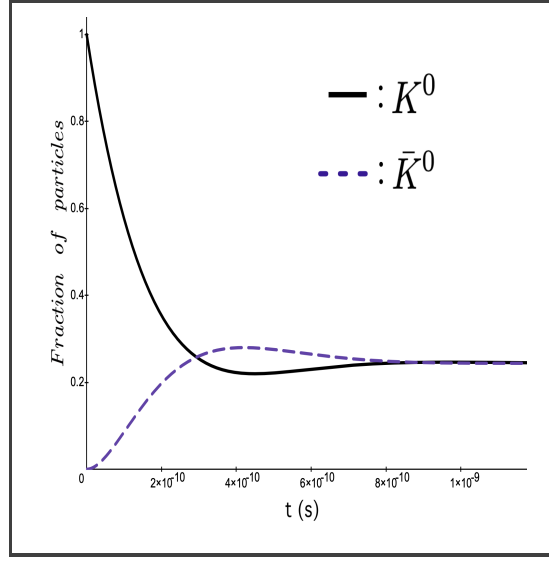


Figure 3.1: Fraction of K^0 and $\bar{K}^0$ as a function of time for an initially pure K^0 beam.

3.1.2 CP Violation

If CP were conserved then, sufficiently far from the K^0 production point, neutral decays to two pions would no longer occur. However, it has been experimentally observed that $K_L^0 \rightarrow \pi^+\pi^-$ occurs with a significant enough rate:

$$\Gamma(K_L^0 \rightarrow \pi^+\pi^-)/\Gamma(K_L^0 \rightarrow \text{all}) \simeq 2 \times 10^{-3} \quad (3.6)$$

There are 2 ways in which CP violations can be accounted for [4]:

1. K_s^0 & K_L^0 are not the exact CP eigenstates. There is mixing of the two states parameterized by ϵ . (Indirect CP violation)
2. Direct CP violation through decay of K_2^0 into both 3π and 2π states, parameterized by ϵ' .

Experimentally it has been found that both the mechanisms do contribute to the measured CP violation, but (1) dominates:

$$\frac{\epsilon'}{\epsilon} = (1.7 \pm 0.3) \times 10^{-3}$$

The first way to account for CP violation is to redefine the mass eigenstates as follows:

$$|K_s\rangle = \frac{1}{\sqrt{1+|\epsilon|^2}}(|K_1^0\rangle + \epsilon|K_2^0\rangle)$$

$$|K_L\rangle = \frac{1}{\sqrt{1+|\epsilon|^2}}(|K_2^0\rangle + \epsilon|K_1^0\rangle)$$

Experimentally it is found $|\epsilon| \sim 2 \times 10^{-3}$.

In the flavour basis which is simply:

$$|K_s\rangle = p|K^0\rangle + q|\bar{K}^0\rangle$$

$$|K_L\rangle = p|K^0\rangle - q|\bar{K}^0\rangle$$

Where p and q are complex numbers satisfying $|p|^2 + |q|^2 = 1$, and for CP violation to occur:

$$\frac{q}{p} \neq 1, \quad \frac{q}{p} = \frac{1 - \epsilon}{1 + \epsilon} \simeq 1 - 2\epsilon$$

Starting again from an initial pure beam of K^0 particles:

$$|\psi(0)\rangle = \frac{1}{2p}(|K_s\rangle + |K_L\rangle) \quad (3.7)$$

Applying the time evolution on this state yields:

$$|\psi(t)\rangle = \frac{1}{2p}(e^{-i\lambda_s t} |K_s\rangle + e^{-i\lambda_L t} |K_L\rangle) \quad (3.8)$$

Substituting back into flavour basis:

$$|\psi(t)\rangle = \frac{1}{2}(e^{-i\lambda_s t} + e^{-i\lambda_L t}) |K^0\rangle + \frac{q}{2p}(e^{-i\lambda_s t} - e^{-i\lambda_L t}) |\bar{K}^0\rangle \quad (3.9)$$

Fraction of K^0 at time t :

$$\begin{aligned} F(K^0) &= |\langle K^0 | \psi(t) \rangle|^2 \\ &= \frac{1}{4}[e^{-\Gamma_s t} + e^{-\Gamma_L t} + 2 \cos(\Delta m t) e^{-\Gamma t}] \end{aligned} \quad (3.10)$$

Fraction of $\bar{K}^0$ at time t :

$$\begin{aligned} F(\bar{K}^0) &= |\langle \bar{K}^0 | \psi(t) \rangle|^2 \\ &= \left| \frac{q}{2p} \right|^2 [e^{-\Gamma_s t} + e^{-\Gamma_L t} - 2 \cos(\Delta m t) e^{-\Gamma t}] \end{aligned} \quad (3.11)$$

But as it is the CP eigenstates that decay into pions, writing the time evolution state in this basis:

$$|\psi(t)\rangle = \frac{1}{\sqrt{2}(1 + \epsilon)}[(e^{-i\lambda_s t} + \epsilon e^{-i\lambda_L t}) |K_1^0\rangle + (e^{-i\lambda_L t} + \epsilon e^{-i\lambda_s t}) |K_2^0\rangle] \quad (3.12)$$

Thus,

$$\begin{aligned} \Gamma(K \rightarrow \pi\pi) &\propto |\langle K_1^0 | \psi(t) \rangle|^2 \\ &= \frac{1}{2} \left| \frac{1}{1 + \epsilon} \right|^2 |(e^{-i\lambda_s t} + \epsilon e^{-i\lambda_L t})|^2 \end{aligned} \quad (3.13)$$

Since $|\epsilon| \ll 1$, and writing ϵ as $|\epsilon|e^{i\phi}$

$$\Gamma(K \rightarrow \pi\pi) = \frac{1}{2}(1 - 2\Re(\epsilon))N_{\pi\pi}[e^{-\Gamma_s t} + |\epsilon|^2 e^{-\Gamma_L t} + 2|\epsilon|e^{-\Gamma t} \cos(\Delta m t - \phi)] \quad (3.14)$$

Where $N_{\pi\pi}$ is a normalization constant.

After several K_s lifetimes only the K_L component remains, that is the CP violating decays:

$$\Gamma(K \rightarrow \pi\pi) \rightarrow \frac{1}{2}(1 - 2\Re(\epsilon))N_{\pi\pi} \cdot |\epsilon|^2 e^{-\Gamma_L t}$$

Similarly if we had started with a pure $\bar{K}^0$ beam then we would have just flipped the sign of the interference term in the expression:

$$\Gamma(\bar{K} \rightarrow \pi\pi) = \frac{1}{2}(1 + 2\Re(\epsilon))N_{\pi\pi}[e^{-\Gamma_s t} + |\epsilon|^2 e^{-\Gamma_L t} - 2|\epsilon|e^{-\Gamma t} \cos(\Delta m t - \phi)] \quad (3.15)$$

Using $\Gamma(K \rightarrow \pi\pi)$ and $\Gamma(\bar{K} \rightarrow \pi\pi)$ we can calculate $|\epsilon|$, which has been experimentally found to be $|\epsilon| = (2.264 \pm 0.035) \times 10^{-3}$ by the CPLEAR Experiment [5]. The experiment also found the value of $\phi = (43.19 \pm 0.73)^\circ$. Using these values equation 3.14 & 3.15 have been plotted as a function of time in figure 3.2.

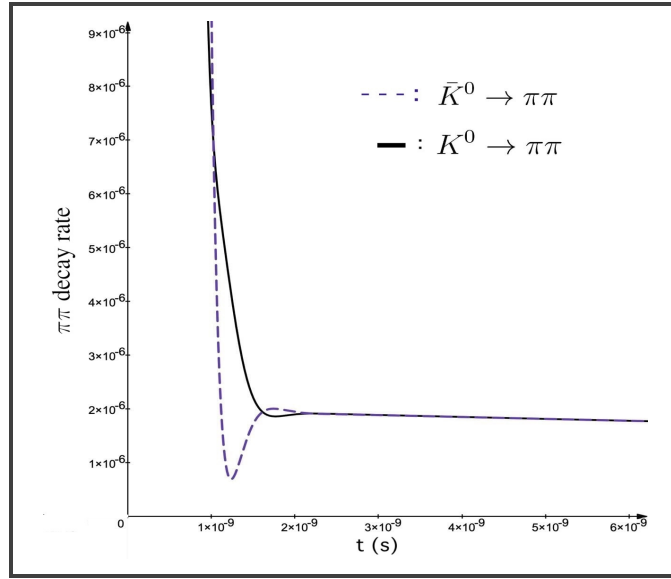


Figure 3.2: $\pi\pi$ decay rate vs time plot for initial pure K^0 and $\bar{K}^0$ beams.

3.2 Neutrino Oscillations

Neutrinos are electrically neutral, nearly massless fermions that interact only via the weak nuclear force and gravity. They were first postulated by Wolfgang Pauli in 1930 to preserve conservation of energy and angular momentum in beta decay process. The neutrino remained hypothetical until its experimental detection by Cowan and Reines in 1956 through inverse beta decay using a nuclear reactor as a source of antineutrinos [6].

Neutrinos come in three known flavors: electron (ν_e), muon (ν_μ), and tau (ν_τ), each associated with a corresponding charged lepton. A fundamental breakthrough in particle physics was the discovery of neutrino oscillation, a quantum mechanical phenomenon where neutrinos change flavor as they propagate. This implied that neutrinos have non-zero mass and that the flavor eigenstates are quantum superpositions of mass eigenstates [7, 8]. It also implied that the associated lepton number is violated. The lepton number violation can occur via $|\Delta L| = 0$ or 2 transitions, where $L = L_e + L_\mu + L_\tau$.

3.2.1 $\Delta L = 0$ Oscillations

(For simplicity we will consider only electron and muon neutrino oscillation)

By definition neutrino state are produced along a lepton.

$$\nu_e, \nu_\mu, \nu_\tau = \text{weak eigenstates}$$

The weak eigenstates can be re-written in the mass eigenstates (ν_1, ν_2) as a coherent linear combination. The mass eigenstates are the free particle solutions to the wave-equation and will be taken to propagate as plane waves

$$|\nu_1(t)\rangle = |\nu_1\rangle e^{i\vec{p}_1 \cdot \vec{x} - iE_1 t} \quad |\nu_2(t)\rangle = |\nu_2\rangle e^{i\vec{p}_2 \cdot \vec{x} - iE_2 t}$$

The two eigenstates are related by the 2×2 *unitary matrix*:

$$\begin{pmatrix} \nu_e \\ \nu_\mu \end{pmatrix} = \begin{pmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{pmatrix} \begin{pmatrix} \nu_1 \\ \nu_2 \end{pmatrix} \quad (3.16)$$

Inversely,

$$\begin{pmatrix} \nu_1 \\ \nu_2 \end{pmatrix} = \begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix} \begin{pmatrix} \nu_e \\ \nu_\mu \end{pmatrix} \quad (3.17)$$

Suppose at $t = 0$ a neutrino is produced in a pure ν_e state, which can be expressed as:

$$|\psi(0)\rangle = |\nu_e\rangle = \cos \theta |\nu_1\rangle + \sin \theta |\nu_2\rangle \quad (3.18)$$

Applying the time evolution of the mass eigenstates assuming the neutrino is propagating in x direction. Making an approximation $v_i \sim c$. Thus $x = ct = t$:

$$|\psi(x)\rangle = \cos \theta |\nu_1\rangle e^{-i(E_1 t - |\vec{p}_1| x)} + \sin \theta |\nu_2\rangle e^{-i(E_2 t - |\vec{p}_2| x)} \quad (3.19)$$

for $m_i \ll E_i$:

$$\begin{aligned} |\vec{p}_i|^2 &= E_i^2 \left(1 - \frac{m_i^2}{E_i^2} \right) \\ \Rightarrow |\vec{p}_i| &= E_i \left(1 - \frac{m_i^2}{E_i^2} \right)^{\frac{1}{2}} \sim E_i - \frac{m_i^2}{2E_i} \end{aligned}$$

Defining:

$$\phi_i = \frac{m_i^2}{2E_i} x$$

we get:

$$|\psi(x)\rangle = \cos \theta |\nu_1\rangle e^{-i\phi_1} + \sin \theta |\nu_2\rangle e^{-i\phi_2} \quad (3.20)$$

ϕ_i is the phase of the wave for the mass eigenstate i at the distance x from the source. Now moving back to the weak eigenstates:

$$|\psi(x)\rangle = (\cos^2(\theta)e^{-i\phi_1} + \sin^2(\theta)e^{-i\phi_2}) |\nu_e\rangle + (e^{-i\phi_2} - e^{-i\phi_1}) \sin(\theta) \cos(\theta) |\nu_\mu\rangle \quad (3.21)$$

If the masses of $|\nu_1\rangle$, $|\nu_2\rangle$ are same, the mass eigenstates remain in phase, $\phi_1 = \phi_2$, and the state remains $|\nu_e\rangle$. If the masses are different then we have a mixing between the two neutrino flavours. The probability for the initial pure $|\nu_e\rangle$ to convert to $|\nu_\mu\rangle$ is given by:

$$\begin{aligned} P(\nu_e \rightarrow \nu_\mu) &= |\langle \nu_\mu | \psi(x) \rangle|^2 \\ &= \cos^2(\theta) \sin^2(\theta) (2 - 2 \cos(\phi_2 - \phi_1)) \\ &= \sin^2(2\theta) \sin^2\left(\frac{\phi_2 - \phi_1}{2}\right) \end{aligned} \quad (3.22)$$

Writing $\Delta m_{21}^2 = m_2^2 - m_1^2$ and $E_1 \simeq E_2 = E$:

$$P(\nu_e \rightarrow \nu_\mu) = \sin^2(2\theta) \sin^2\left(\frac{\Delta m_{21}^2 x}{4E}\right) \quad (3.23)$$

This is the flavour oscillation probability. The corresponding flavour survival probability is:

$$P(\nu_e \rightarrow \nu_e) = 1 - \sin^2(2\theta) \sin^2\left(\frac{\Delta m_{21}^2 x}{4E}\right) \quad (3.24)$$

Thus the wavelength for oscillation is:

$$\lambda_{osc} = \frac{4\pi E}{\Delta m^2}$$

As the mixing matrix in 2 flavour oscillation is real,

$$P(\nu_e \rightarrow \nu_\mu) = P(\bar{\nu}_e \rightarrow \bar{\nu}_\mu)$$

Thus CP is not violated.

All expressions above are in Natural units, converting to more useful units here gives:

$$\Delta_{21} = 1.27 \frac{\Delta m_{21}^2 (\text{eV}^2) x (\text{Km})}{E (\text{GeV})} \quad \text{and} \quad \lambda_{osc} (\text{Km}) = 2.47 \frac{E (\text{GeV})}{\Delta m^2 (\text{eV}^2)}$$

For atmospheric neutrinos typical values of the parameters taken as follows: $\Delta m^2 = 0.0025 \text{ eV}^2$, $\sin^2(2\theta) = 0.95$, $E_\nu = 1 \text{ GeV}$ the oscillation probability between the flavours of neutrino is plotted against travel distance in plot 3.3. Plugging in the values $\lambda_{osc} = 988 \text{ Km}$.

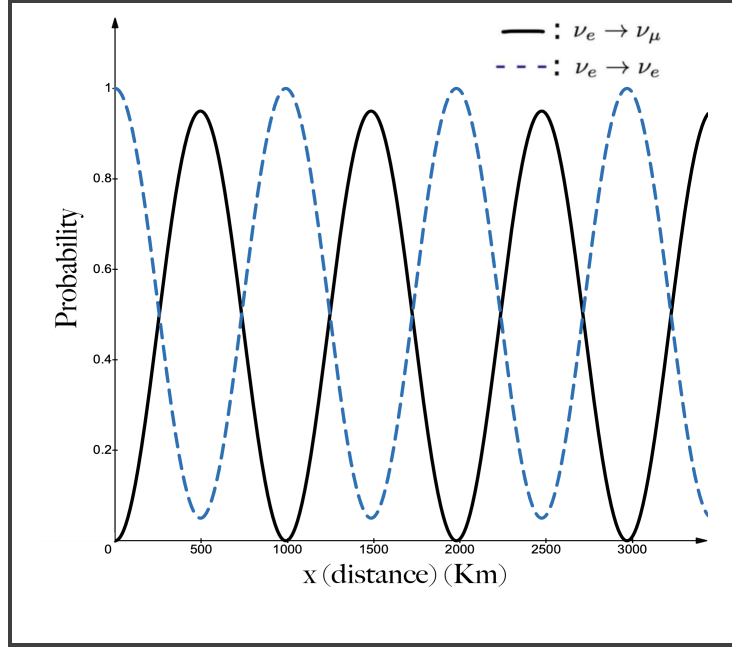


Figure 3.3: 2 flavour neutrino oscillation

For CP violation a complex phase in the mixing matrix is required that can not be removed by rephasing the fields. This only happens in the 3 flavour neutrino system.

3.2.2 3 Flavour neutrino oscillation

To extend to 3 flavours we now require a 3×3 unitary matrix U which is also called the PMNS matrix (Pontecorvo-Maki-Nakagawa-Sakata). It has to be unitary to conserve probability. The two different eigenstates are related via PMNS matrix as follows:

$$\begin{pmatrix} \nu_e \\ \nu_\mu \\ \nu_\tau \end{pmatrix} = \begin{pmatrix} U_{e1} & U_{e2} & U_{e3} \\ U_{\mu1} & U_{\mu2} & U_{\mu3} \\ U_{\tau1} & U_{\tau2} & U_{\tau3} \end{pmatrix} \begin{pmatrix} \nu_1 \\ \nu_2 \\ \nu_3 \end{pmatrix} \quad (3.25)$$

$$\text{As } U^\dagger U = I \implies U^{-1} = U^\dagger = (U^*)^T$$

So the inverse relation can be written as:

$$\begin{pmatrix} \nu_1 \\ \nu_2 \\ \nu_3 \end{pmatrix} = \begin{pmatrix} U_{e1}^* & U_{\mu1}^* & U_{\tau1}^* \\ U_{e2}^* & U_{\mu2}^* & U_{\tau2}^* \\ U_{e3}^* & U_{\mu3}^* & U_{\tau3}^* \end{pmatrix} \begin{pmatrix} \nu_e \\ \nu_\mu \\ \nu_\tau \end{pmatrix} \quad (3.26)$$

Lets start with a pure $|\nu_e\rangle$ state as before, at $t = 0$:

$$|\psi(0)\rangle = U_{e1} |\nu_1\rangle + U_{e2} |\nu_2\rangle + U_{e3} |\nu_3\rangle \quad (3.27)$$

Applying the time evolution on the mass eigenstates (assuming direction of propagation to be x):

$$|\psi(t)\rangle = U_{e1} |\nu_1\rangle e^{-i(E_1 t - |\vec{p}_1| x)} + U_{e2} |\nu_2\rangle e^{-i(E_2 t - |\vec{p}_2| x)} + U_{e3} |\nu_3\rangle e^{-i(E_3 t - |\vec{p}_3| x)} \quad (3.28)$$

Just like before we can approximate

$$\phi_i \sim \frac{m_i^2}{2E_i} x$$

$$\implies |\psi(x)\rangle = U_{e1} |\nu_1\rangle e^{-i\phi_1} + U_{e2} |\nu_2\rangle e^{-i\phi_2} + U_{e3} |\nu_3\rangle e^{-i\phi_3} \quad (3.29)$$

Writing the equation 3.29 in terms of weak eigenstates

$$\begin{aligned} |\psi(x)\rangle &= U_{e1} e^{-i\phi_1} (U_{e1}^* |\nu_e\rangle + U_{\mu 1}^* |\nu_\mu\rangle + U_{\tau 1}^* |\nu_\tau\rangle) \\ &\quad + U_{e2} e^{-i\phi_2} (U_{e2}^* |\nu_e\rangle + U_{\mu 2}^* |\nu_\mu\rangle + U_{\tau 2}^* |\nu_\tau\rangle) \\ &\quad + U_{e3} e^{-i\phi_3} (U_{e3}^* |\nu_e\rangle + U_{\mu 3}^* |\nu_\mu\rangle + U_{\tau 3}^* |\nu_\tau\rangle) \end{aligned} \quad (3.30)$$

Rearrangement gives

$$\begin{aligned} |\psi(x)\rangle &= (U_{e1} U_{e1}^* e^{-i\phi_1} + U_{e2} U_{e2}^* e^{-i\phi_2} + U_{e3} U_{e3}^* e^{-i\phi_3}) |\nu_e\rangle \\ &\quad + (U_{e1} U_{\mu 1}^* e^{-i\phi_1} + U_{e2} U_{\mu 2}^* e^{-i\phi_2} + U_{e3} U_{\mu 3}^* e^{-i\phi_3}) |\nu_\mu\rangle \\ &\quad + (U_{e1} U_{\tau 1}^* e^{-i\phi_1} + U_{e2} U_{\tau 2}^* e^{-i\phi_2} + U_{e3} U_{\tau 3}^* e^{-i\phi_3}) |\nu_\tau\rangle \end{aligned} \quad (3.31)$$

The probability of oscillation into ν_μ at distance x from source of pure ν_e becomes:

$$\begin{aligned} P(\nu_e \rightarrow \nu_\mu) &= |\langle \nu_\mu | \psi(x) \rangle|^2 \\ &= |(U_{e1} U_{\mu 1}^* e^{-i\phi_1} + U_{e2} U_{\mu 2}^* e^{-i\phi_2} + U_{e3} U_{\mu 3}^* e^{-i\phi_3})|^2 \\ &= |U_{e1} U_{\mu 1}^*|^2 + |U_{e2} U_{\mu 2}^*|^2 + |U_{e3} U_{\mu 3}^*|^2 \\ &\quad + 2\Re(U_{e1} U_{\mu 2} U_{e2}^* U_{\mu 1}^* e^{-i(\phi_1 - \phi_2)} + U_{e1} U_{\mu 3} U_{e3}^* U_{\mu 1}^* e^{-i(\phi_1 - \phi_3)} + U_{e2} U_{\mu 3} U_{e3}^* U_{\mu 2}^* e^{-i(\phi_2 - \phi_3)}) \end{aligned} \quad (3.32)$$

From the properties of the *PMNS* matrix:

$$\begin{aligned} U_{e1} U_{\mu 1}^* + U_{e2} U_{\mu 2}^* + U_{e3} U_{\mu 3}^* &= 0 \\ |U_{e1} U_{\mu 1}^* + U_{e2} U_{\mu 2}^* + U_{e3} U_{\mu 3}^*|^2 &= 0 \\ \implies |U_{e1} U_{\mu 1}^*|^2 + |U_{e2} U_{\mu 2}^*|^2 + |U_{e3} U_{\mu 3}^*|^2 &= -2\Re(U_{e1} U_{\mu 2} U_{e2}^* U_{\mu 1}^* + U_{e1} U_{\mu 3} U_{e3}^* U_{\mu 1}^* + U_{e2} U_{\mu 3} U_{e3}^* U_{\mu 2}^*) \end{aligned}$$

Simplifying equation 3.32:

$$\begin{aligned} P(\nu_e \rightarrow \nu_\mu) &= 2\Re\{U_{e1} U_{\mu 2} U_{e2}^* U_{\mu 1}^* [e^{-i(\phi_1 - \phi_2)} - 1]\} \\ &\quad + 2\Re\{U_{e1} U_{\mu 3} U_{e3}^* U_{\mu 1}^* [e^{-i(\phi_1 - \phi_3)} - 1]\} \\ &\quad + 2\Re\{U_{e2} U_{\mu 3} U_{e3}^* U_{\mu 2}^* [e^{-i(\phi_2 - \phi_3)} - 1]\} \end{aligned} \quad (3.33)$$

Similarly the expression for survival probability of $|\nu_e\rangle$ is:

$$\begin{aligned} P(\nu_e \rightarrow \nu_e) &= |\langle \nu_e | \psi(x) \rangle|^2 \\ &= |U_{e1} U_{e1}^* e^{-i\phi_1} + U_{e2} U_{e2}^* e^{-i\phi_2} + U_{e3} U_{e3}^* e^{-i\phi_3}|^2 \\ &= 1 + 2|U_{e1}|^2 |U_{e2}|^2 \Re\{[e^{-i(\phi_1 - \phi_2)} - 1]\} \\ &\quad + 2|U_{e1}|^2 |U_{e3}|^2 \Re\{[e^{-i(\phi_1 - \phi_3)} - 1]\} \\ &\quad + 2|U_{e3}|^2 |U_{e2}|^2 \Re\{[e^{-i(\phi_2 - \phi_3)} - 1]\} \end{aligned}$$

Define:

$$\Delta_{ij} = \frac{\Delta m_{ij}^2}{4E} x$$

Thus

$$P(\nu_e \rightarrow \nu_e) = 1 - 4|U_{e1}|^2|U_{e2}|^2 \sin^2(\Delta_{21}) - 4|U_{e1}|^2|U_{e3}|^2 \sin^2(\Delta_{31}) - 4|U_{e3}|^2|U_{e2}|^2 \sin^2(\Delta_{31}) \quad (3.34)$$

Similar expressions can be obtained for the muon and the tau neutrino survival probabilities. Since there are only 3 neutrino flavours there are only 2 independent mass-squared differences, i.e.

$$\Delta m_{31}^2 = (\Delta m_{32}^2 - \Delta m_{21}^2)$$

And in the above equations only two of the Δ_{ij} are independent [9].

Neutrino oscillations require non zero neutrino masses. Neutrino mass hasn't been directly measured but a limit on the mass is determined:

$$m_\nu(e) < 2 \text{ eV} ; m_\nu(\mu) < 0.17 \text{ MeV} ; m_\nu(\tau) < 18.2 \text{ MeV}$$

3.2.3 CP Violation

Previously derived equation 3.33 provides the probability for $\nu_e \rightarrow \nu_\mu$:

$$\begin{aligned} P(\nu_e \rightarrow \nu_\mu) = & 2\Re\{U_{e1}U_{\mu2}U_{e2}^*U_{\mu1}^*[e^{-i(\phi_1-\phi_2)} - 1]\} \\ & + 2\Re\{U_{e1}U_{\mu3}U_{e3}^*U_{\mu1}^*[e^{-i(\phi_1-\phi_3)} - 1]\} \\ & + 2\Re\{U_{e2}U_{\mu3}U_{e3}^*U_{\mu2}^*[e^{-i(\phi_2-\phi_3)} - 1]\} \end{aligned}$$

The oscillation probability for $\nu_\mu \rightarrow \nu_e$ obtained similarly is:

$$\begin{aligned} P(\nu_\mu \rightarrow \nu_e) = & 2\Re\{U_{e2}U_{\mu1}U_{e1}^*U_{\mu2}^*[e^{-i(\phi_1-\phi_2)} - 1]\} \\ & + 2\Re\{U_{e3}U_{\mu1}U_{e1}^*U_{\mu3}^*[e^{-i(\phi_1-\phi_3)} - 1]\} \\ & + 2\Re\{U_{e3}U_{\mu2}U_{e2}^*U_{\mu3}^*[e^{-i(\phi_2-\phi_3)} - 1]\} \end{aligned} \quad (3.35)$$

Unless the elements of $PMNS$ matrix are real:

$$P(\nu_e \rightarrow \nu_\mu) \neq P(\nu_\mu \rightarrow \nu_e)$$

The effects of T,CP and CPT on neutrino oscillations [9]:

$$\nu_e \rightarrow \nu_\mu \xrightarrow{\hat{T}} \nu_\mu \rightarrow \nu_e$$

$$\nu_e \rightarrow \nu_\mu \xrightarrow{\hat{C}\hat{P}} \bar{\nu}_e \rightarrow \bar{\nu}_\mu$$

$$\nu_e \rightarrow \nu_\mu \xrightarrow{\hat{C}\hat{P}\hat{T}} \bar{\nu}_\mu \rightarrow \bar{\nu}_e$$

If weak interactions is invariant under CPT

$$P(\nu_e \rightarrow \nu_\mu) = P(\bar{\nu}_\mu \rightarrow \bar{\nu}_e)$$

and,

$$P(\nu_\mu \rightarrow \nu_e) = P(\bar{\nu}_e \rightarrow \bar{\nu}_\mu)$$

If the $PMNS$ matrix is not purely real, then

$$P(\nu_e \rightarrow \nu_\mu) \neq P(\nu_\mu \rightarrow \nu_e)$$

and,

$$P(\nu_e \rightarrow \nu_\mu) \neq P(\bar{\nu}_e \rightarrow \bar{\nu}_\mu) \quad (3.36)$$

Therefore, CP is violated in neutrino oscillation.

3.2.4 PMNS Matrix

The $PMNS$ matrix is usually expressed in terms of 3 rotation angles ($\theta_{12}, \theta_{23}, \theta_{13}$) and a complex phase (δ), using the following notation

$$s_{ij} = \sin(\theta_{ij}), \quad c_{ij} = \cos(\theta_{ij})$$

The $PMNS$ matrix can be decomposed as:

$$U_{PMNS} = \begin{pmatrix} U_{e1} & U_{e2} & U_{e3} \\ U_{\mu 1} & U_{\mu 2} & U_{\mu 3} \\ U_{\tau 1} & U_{\tau 2} & U_{\tau 3} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & c_{23} & s_{23} \\ 0 & -s_{23} & c_{23} \end{pmatrix} \cdot \begin{pmatrix} c_{13} & 0 & s_{13}e^{-i\delta} \\ 0 & 1 & 0 \\ -s_{13}e^{i\delta} & 0 & c_{13} \end{pmatrix} \cdot \begin{pmatrix} c_{12} & s_{12} & 0 \\ -s_{12} & c_{12} & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (3.37)$$

$$U_{PMNS} = \begin{pmatrix} c_{12}c_{13} & c_{13}s_{12} & s_{13}e^{-i\delta} \\ -s_{12}s_{23} - c_{12}s_{23}s_{13}e^{i\delta} & c_{12}c_{23} - s_{12}s_{23}s_{13}e^{i\delta} & s_{23}c_{13} \\ s_{12}s_{23} - c_{12}c_{23}s_{13}e^{i\delta} & -c_{12}s_{23} - s_{12}s_{13}c_{23}e^{i\delta} & c_{23}c_{13} \end{pmatrix} \quad (3.38)$$

- θ_{12} : governs solar oscillations
- θ_{23} : governs atmospheric oscillations
- θ_{13} : couples ν_e to ν_3
- δ : CP violating Dirac phase

3.3 Contrast between quark and lepton sector

CP violation is observed in both the quark and the lepton sector only when there are 3 flavours. In case of 2 flavours (2×2 mixing matrix) CP violation is not possible. Even though there are only 2 quarks in the neutral kaon system the mixing matrix is written for 3 quark flavours and thus CP violation is observed. The following table contrasts between the two systems we have studied in the previous sections.

Feature	Kaon System	Neutrino System
Particle Types	Neutral Kaons: $K^0, \bar{K}^0$	Neutrinos: ν_e, ν_μ, ν_τ
Interactions	Strong & Weak	Weak
Decay	Decay into 3 pions or 2 pions state	Do not decay
Oscillation	$ K^0\rangle \rightarrow \bar{K}^0\rangle$	$\nu_e \rightarrow \nu_\mu$
Mixing Matrix	CKM	PMNS
CP violation	$K_L^0 \rightarrow 2$ pions and Asymmetric decay rates	$P(\nu_e \rightarrow \nu_\mu) \neq P(\bar{\nu}_e \rightarrow \bar{\nu}_\mu)$
Source of CP violation	CP violating Dirac phase in the mixing matrix	
Intensity of violation	Very less (about 0.1%)	Large CP violation?

Table 3.1: Comparison of CP violation in the quark and lepton sectors

Chapter 4

Quantum Computation

The emerging technology of quantum computers will help us understand quantum mechanical phenomena better than the classical counterparts, because a quantum computer unlike a classical computer inherently runs on quantum states and thus gives a better formalism to study the quantum effects. In this chapter we get a short overview of the theory behind few of the basic operations of a quantum computer.

4.1 Qubits

Qubits are treated as abstract mathematical objects, which are realized as actual physical systems. Just like a classical bit has a state - either 0 or 1 - a qubit also has a base state either $|0\rangle$ or $|1\rangle$. The difference between bits and qubits is that a qubit can be in a state other than $|0\rangle$ or $|1\rangle$. It is also possible to form linear combinations of states, called superposition [10]:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle \quad (4.1)$$

The numbers α and β are complex numbers. Thus the state of a qubit is a unit vector in a two-dimensional complex vector space with $|0\rangle$ and $|1\rangle$ as its orthonormal basis which are also known as computational basis states. This corresponds to the fact that $|\alpha|^2 + |\beta|^2 = 1$. $|\alpha|^2$ and $|\beta|^2$ are the probabilities of the measurement of the qubit giving the result $|0\rangle$ and $|1\rangle$ respectively. This means that a qubit can exist in a continuum of states between $|0\rangle$ and $|1\rangle$ until it is observed. **Bloch sphere** visualization of a qubit can be made by considering a 3-dimensional unit sphere made by considering the angles θ and ϕ that define a point on it. Thus the state of a qubit can be represent as:

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right) |0\rangle + e^{i\phi} \sin\left(\frac{\theta}{2}\right) |1\rangle \quad (4.2)$$

4.2 2 Qubit System

For 2 qubit systems the computational basis states are $|00\rangle$, $|01\rangle$, $|10\rangle$ & $|11\rangle$. A pair of qubit can also exist in superposition of these 4 states.

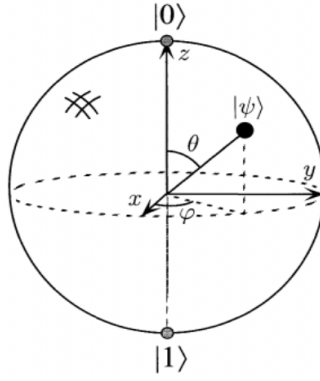


Figure 4.1: Bloch sphere representation of a qubit

Source: fig 1.3, Quantum Computation and quantum information, [10]

4.3 Quantum Gates

To preserve probabilities quantum gates have to be describable by unitary matrices. Any unitary matrix acting on the qubit is a valid quantum gate. In general terms a quantum gate acting on a state of the given qubit system can transform it into another possible state of that system. There are few gates which are very useful to our study which are:

Hadamard Gate:

$$H \equiv \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Not gate:

$$X \equiv \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

U gate:

$$U(\theta, \lambda, \phi) = \begin{bmatrix} \cos(\theta/2) & -e^{i\lambda} \sin(\theta/2) \\ e^{i\phi} \sin(\theta/2) & e^{i(\phi+\lambda)} \cos(\theta/2) \end{bmatrix}$$

The **Hadamard gate (H)** is a very crucial gate which acts on the 2 basis states and outputs a state that has a 50% probability to yield either of the basis states.

$$H |0\rangle = \frac{1}{\sqrt{2}}[|0\rangle + |1\rangle], \quad H |1\rangle = \frac{1}{\sqrt{2}}[|0\rangle - |1\rangle]$$

Mean while the **Not gate (X)** is used to switch the basis states from one to the other.

$$X |0\rangle = |1\rangle$$

The **U gate** or *Unitary gate* is the general representation of any arbitrary 2×2 quantum gate. It rotates the statevector on the bloch sphere parameterized by θ , ϕ and λ .

4.3.1 Multi Qubit Gates

Multi qubit gates refer to those quantum gates that take more than one qubit as input, eg: Controlled gates. Controlled gates have 2 input qubits, known as the *control qubit* and the *target qubit*. A controlled- U gate will operate U on the target qubit only if the control qubit is in state $|1\rangle$.

4.3.2 Measurement of a State

When we measure the state of a qubit in arbitrary superposition of the basis states, we get one of the basis as output. The process of measurement collapses the superposition and we get a state that becomes a classical bit equivalent. The coefficients of the basis states before measurement determine what's the probability for measuring that particular basis as output.

4.4 Quantum Circuit

A quantum circuit consists of qubit registers and classical registers. These can carry a qubit or a classical bit respectively. We can apply respectively quantum and classical gates on these registers. Generally when drawing a quantum circuit the gates are drawn on a specific qubit from left to right. The leftmost gate will act on the qubit first and the transformed state then the next gate is acted on it. Due to the nature of qubits there are few restrictions on what kind of circuit can be constructed.

- There are no loops in quantum circuit.
- *Fanin* is not allowed. Classically fanin is 'bit wise OR'.
- *Fanout* is not allowed. Classically fanout is copying a bit.
- **No cloning theorem** in quantum computing doesn't allow copying of a qubit of unknown state.

4.5 Bell States

Bell states or the EPR states or EPR pairs, after some of the people - Bell, and Einstein, Podolsky, and Rosen - who first pointed out the properties of states like these, are the outputs of a quantum circuit given in figure 4.2 for different possible input. Bell states are a set of maximally

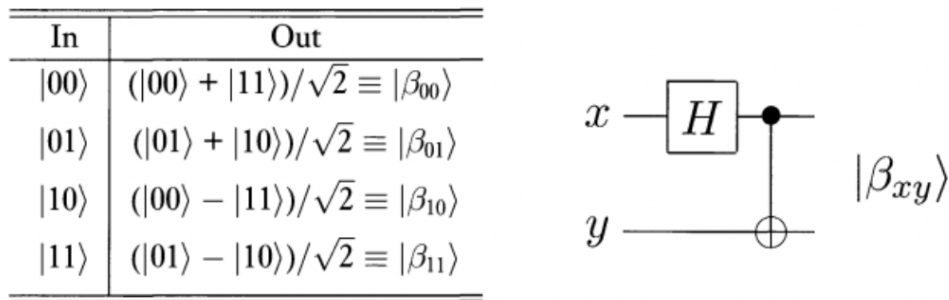


Figure 4.2: Bell state

Source: Fig 1.12, Quantum Computing and Quantum Information [10]

entangled two-qubit quantum states with the property that measurement of one qubit instantly determines the outcome of the other qubit. Bell states have a lot of applications in various different areas. In our simulation we will use bell states to create an initial mixture of equal K_s & K_L .

Chapter 5

Qiskit Simulation for Kaon Decay

In this chapter we use the Qiskit module from IBM Quantum Initiative to create quantum circuits to simulate the phenomena of Kaon decay discussed in chapter 3.

5.1 Kaon System Simulation (No CP violation)

We saw that without considering CP violation we had two processes happening on the initial pure beam of K_0 particles which were re-expressed in terms of the CP eigenstates K_s & K_L . We had K_s decaying into 2π and K_L decaying into 3π with different decay rates.

5.1.1 Defining States

To simulate the time evolution of the K_0 beam we go to the CP basis. To account for the decays of kaons we take a 2 qubit system and define the basis states as follows:

- $|00\rangle$: K_s
- $|01\rangle$: $\pi\pi$ [This state is to denote the number of K_s decays to $\pi\pi$]
- $|11\rangle$: K_L
- $|10\rangle$: $\pi\pi\pi$ [This state is to denote the number of K_L decays to $\pi\pi\pi$]

The key to simulate Kaon decay in this basis is to allow state transformations between $|00\rangle \rightarrow |01\rangle$ and $|11\rangle \rightarrow |10\rangle$ through the U-Gate of appropriate parameter.

5.2 Procedure

1. We begin with a state $\frac{1}{\sqrt{2}}[|00\rangle + |11\rangle]$ which is an analog of K_0 beam. Note that this is also the bell state $|\beta_{00}\rangle$ from figure 4.2.
2. We then time evolve the state and apply the transformation: $|00\rangle \rightarrow \alpha_1 |00\rangle + \alpha_2 |01\rangle$ and $|11\rangle \rightarrow \beta_1 |11\rangle + \beta_2 |10\rangle$. Where α_i and β_i are time dependent coefficients determined using equation 3.3.

3. A special condition that we need to incorporate into the simulation is that α_2, β_2 are strictly increasing with time. This is to incorporate the fact that once a K_s or K_L decays into corresponding Pion state it can not transition back to the Kaon state.
4. If we measure both the qubits after applying the above state transformations then we should be able to simulate the Kaon decay process and obtain the counts of each of the possible final states.

5.3 Algorithm: Kaon Decay

1. **Initialize:** $|q_0\rangle |q_1\rangle = |0\rangle |0\rangle$
2. **Create Bell State:** $|\beta_{00}\rangle$: $\hat{H} |q_0\rangle$ followed by CX (controlled not) gate on $|q_1\rangle$ with $|q_0\rangle$ as control $\rightarrow \frac{1}{\sqrt{2}}[|00\rangle + |11\rangle]$.
3. **Apply Time Evolution:** Apply U gates that act as time evolution.
4. **Measure the State:** Measure the state of both the qubits and each measurement the represents the state of the system.

If we follow the above procedure to create a quantum circuit we get fig 5.1

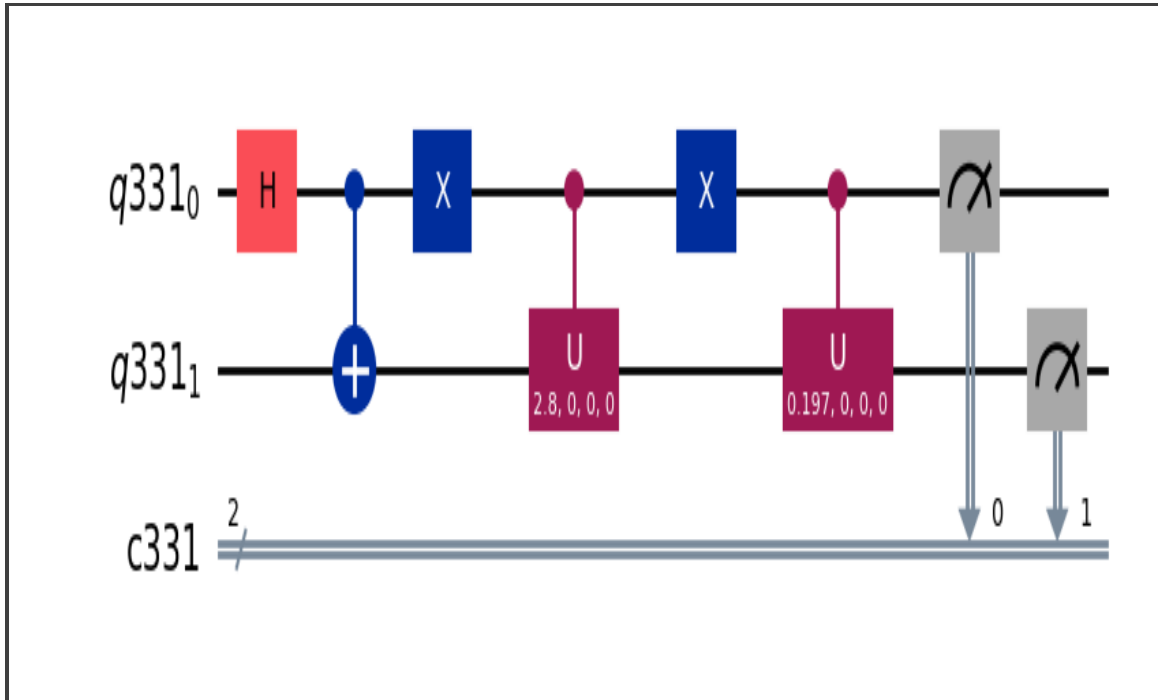


Figure 5.1: Quantum circuit to simulate Kaon decays.

5.4 Defining the U gates

For simplicity we will re-scale the time scales:

$$1 \times 10^{-10} s = 1 \text{ units}$$

$$\implies \Gamma_s = 1.12 \text{ unit}^{-1}, \Gamma_L = 1.93 \times 10^{-3} \text{ unit}^{-1}$$

Lets look at what happens with the K_s component as we apply time evolution.

$$|K_s\rangle \rightarrow e^{-\frac{\Gamma_s t}{2}} |K_s\rangle + \alpha |\pi\pi\rangle$$

$$\implies \alpha^2 = 1 - e^{-\Gamma_s t}$$

We now need to write a U-gate that performs this rotation for time t . We need to find θ such that:

$$U(\theta(t), \lambda, \phi) = \mathbb{U}(t) \implies \mathbb{U}(t) |K_s\rangle = e^{-\frac{\Gamma_s t}{2}} |K_s\rangle + (1 - e^{-\Gamma_s t})^{\frac{1}{2}} |\pi\pi\rangle \quad (5.1)$$

From the definition of U-gate calculate the rotation angle:

$$\cos(\theta/2) = e^{-\Gamma_s t/2}, \quad \sin(\theta/2) = (1 - e^{-\Gamma_s t})^{\frac{1}{2}}$$

For K_s :

$$\theta(t) = 2 \tan^{-1} \left(\frac{(1 - e^{-\Gamma_s t})^{\frac{1}{2}}}{e^{-\Gamma_s t/2}} \right) \quad (5.2)$$

Simplifying equation 5.2:

$$\theta_s = 2 \tan^{-1} \left(\sqrt{t}(1.058 + 0.3t) \right) \quad (5.3)$$

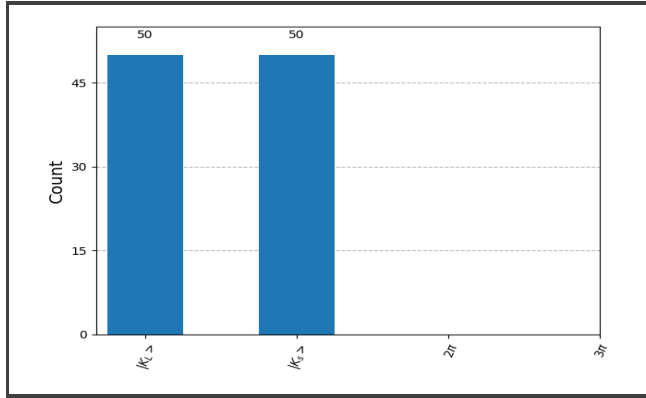
Similarly:

$$\theta_L = 2 \tan^{-1} \left(\sqrt{t}(0.044 + 0.00002t) \right) \quad (5.4)$$

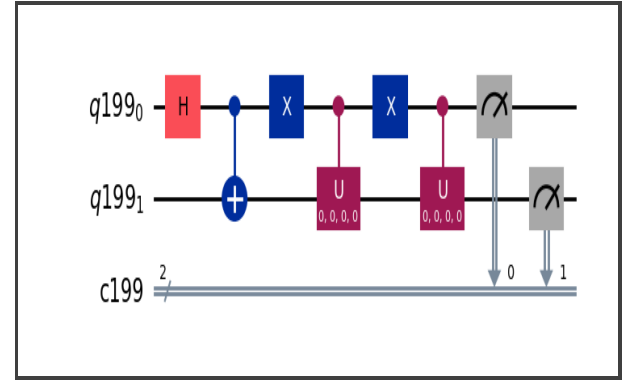
Thus we get 2 gates for K_s and K_L components: $U(\theta_s, 0, 0)$ & $U(\theta_L, 0, 0)$ respectively. These gates are conditionally applied on the second qubit taking the first qubit as control.

5.5 Results

The Aer backend simulator is used to simulate the quantum circuit virtually. Each simulated result is initiated as a pure K^0 beam of 100 particles. Each run of the circuit yields a state as output which is stored in a dictionary called counts. At the end the counts is then plotted. Although we use the appropriate gates, we should be aware that the measurement process in quantum computers are probabilistic. This means having a 50% linear combination of two states doesn't mean that we will get 50 of state 1 and 50 of state 2. In fact we can get a measurement as 35 of state 1 and 45 of state 2. If we increase the shots from 100 to a larger number the result become more accurate. But at 100 shots we can still do a qualitative analysis and see the different decay rates of K_s and K_L . At time t the percentage of each state $\{K_s, K_L, \pi\pi, \pi\pi\pi\}$ is plotted in the histogram:

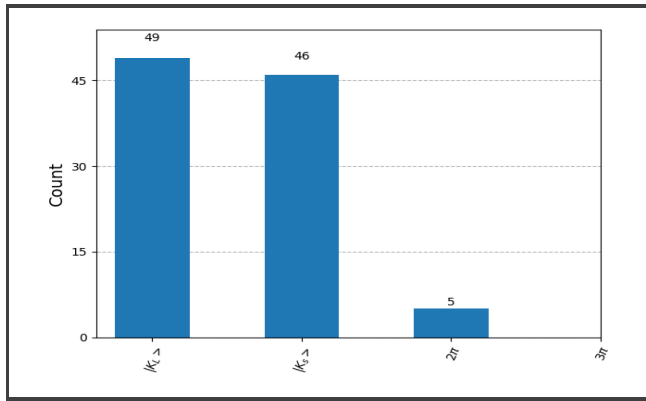


% of particles in each state

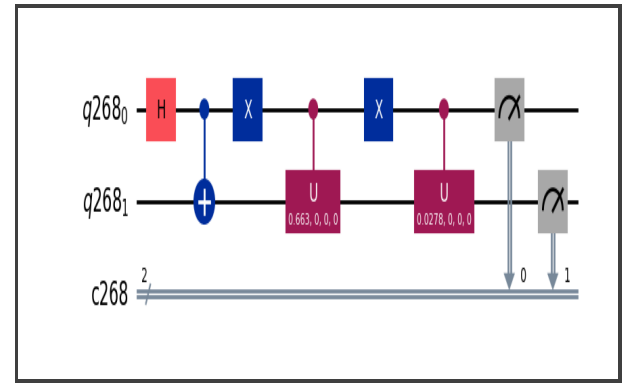


Circuit with appropriate U-gates

Figure 5.2: $t = 0$ units

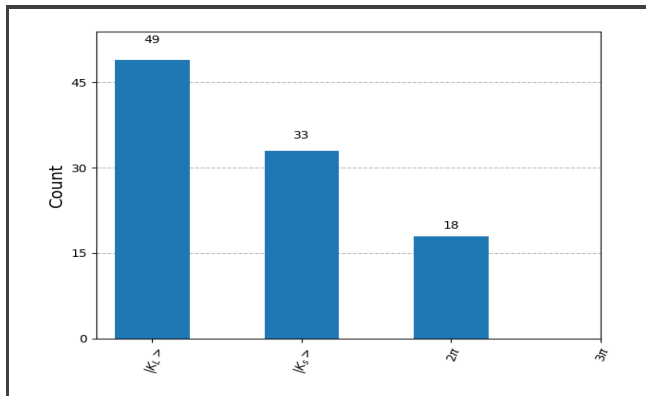


% of particles in each state

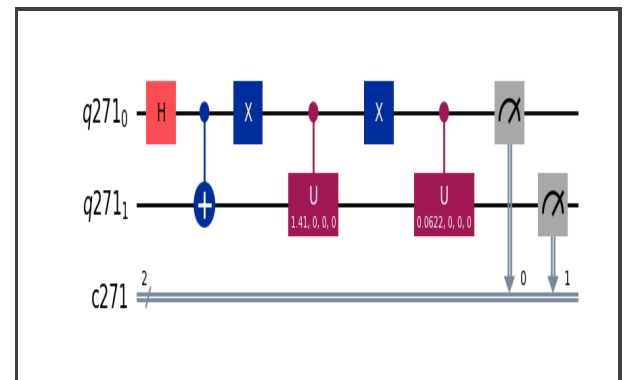


Circuit with appropriate U-gates

Figure 5.3: $t = 0.1$ units

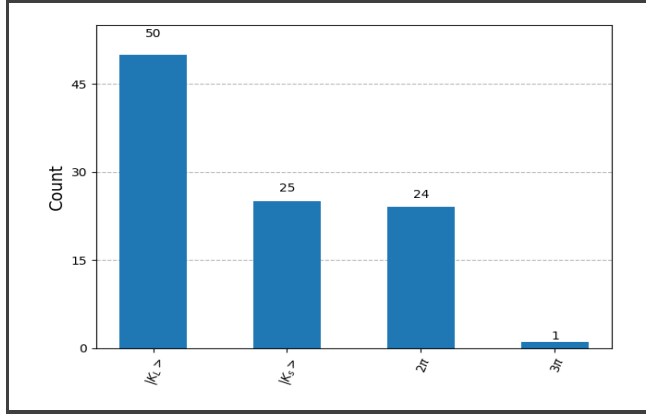


% of particles in each state

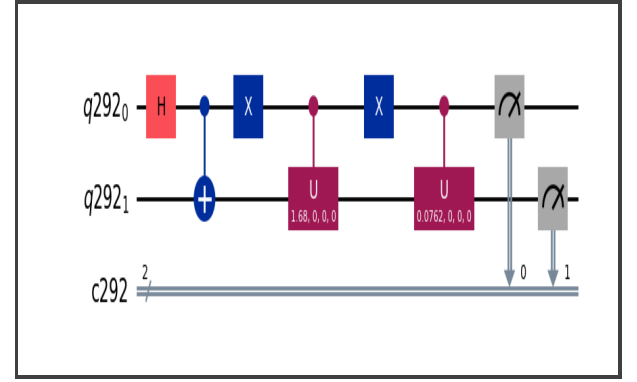


Circuit with appropriate U-gates

Figure 5.4: $t = 0.5$ units

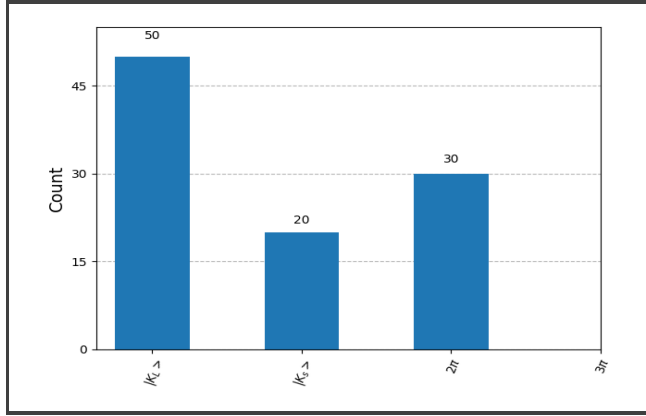


% of particles in each state

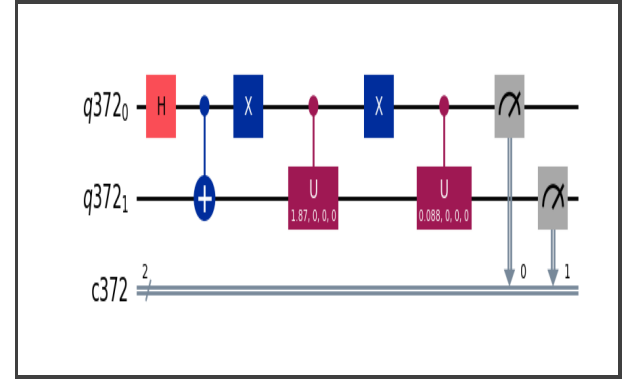


Circuit with appropriate U-gates

Figure 5.5: $t = 0.75$ units

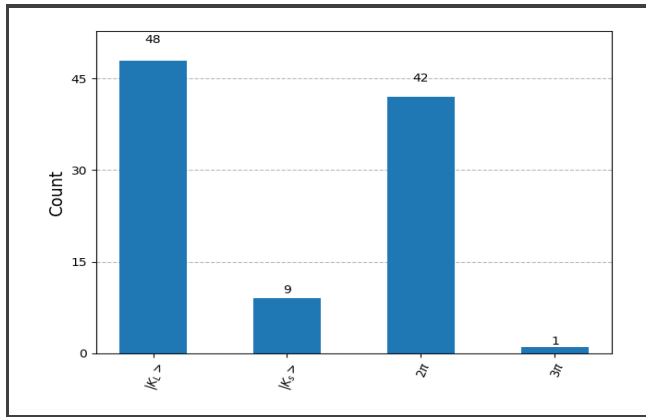


% of particles in each state

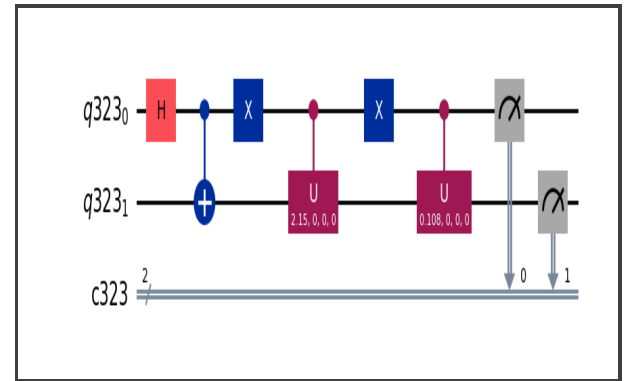


Circuit with appropriate U-gates

Figure 5.6: $t = 1$ units

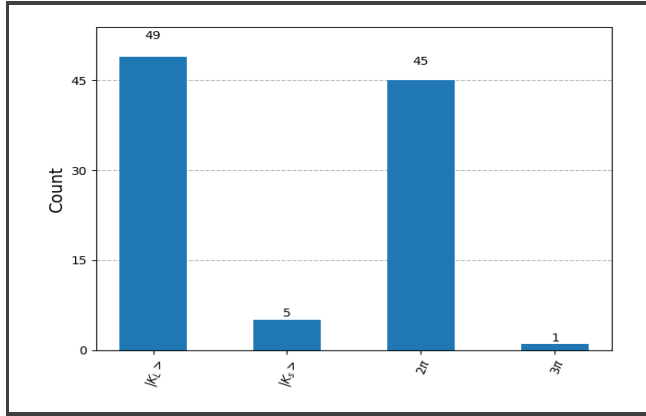


% of particles in each state

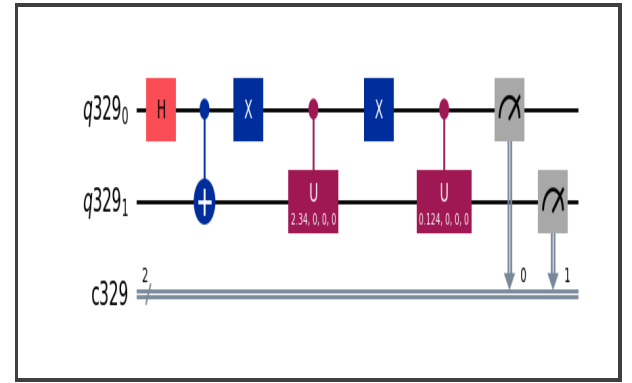


Circuit with appropriate U-gates

Figure 5.7: $t = 1.5$ units

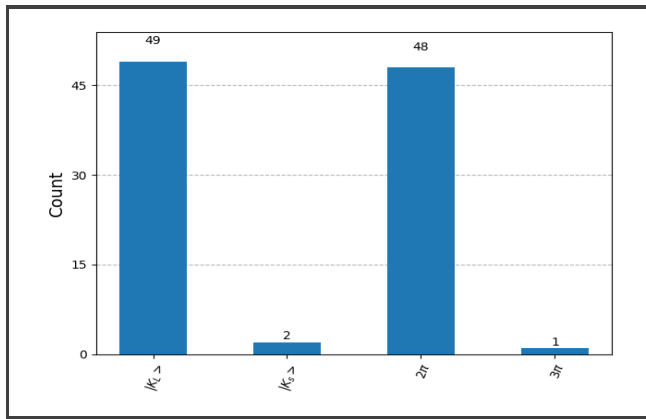


% of particles in each state

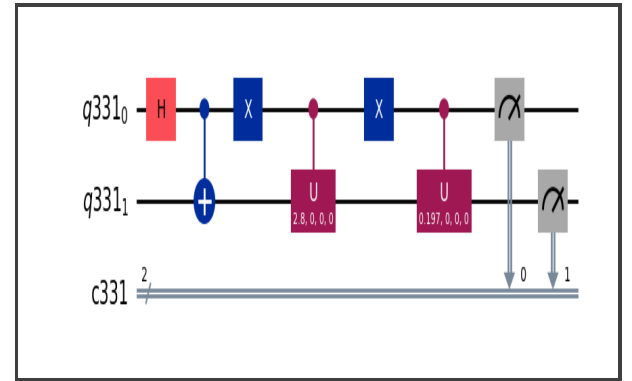


Circuit with appropriate U-gates

Figure 5.8: $t = 2$ units

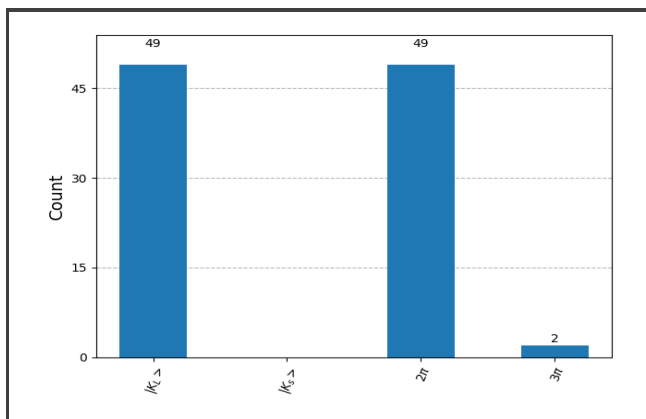


% of particles in each state

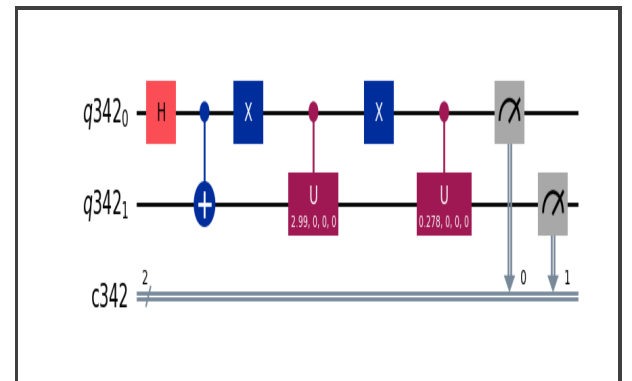


Circuit with appropriate U-gates

Figure 5.9: $t = 5$ units

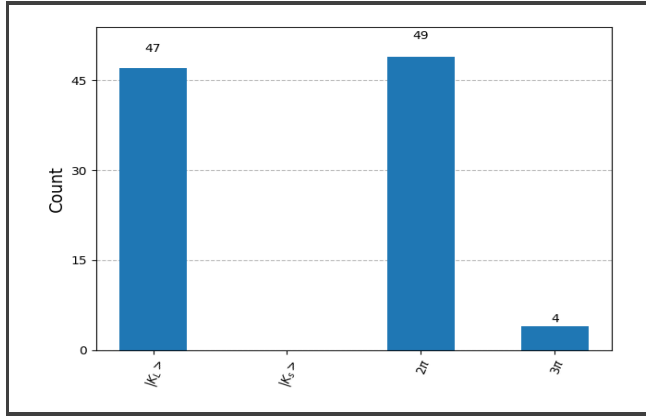


% of particles in each state

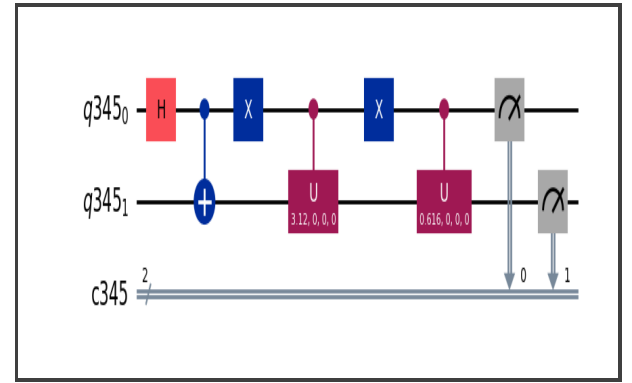


Circuit with appropriate U-gates

Figure 5.10: $t = 10$ units

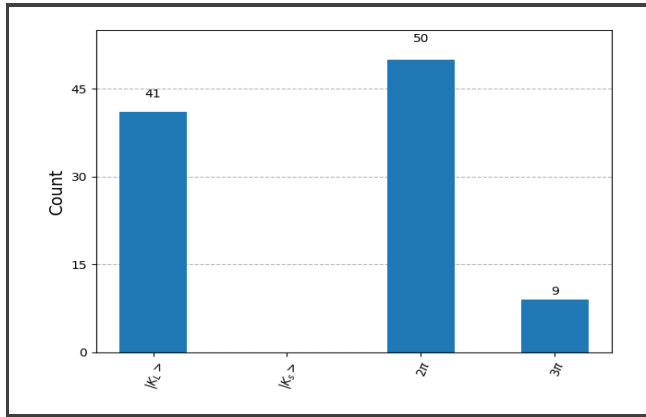


% of particles in each state

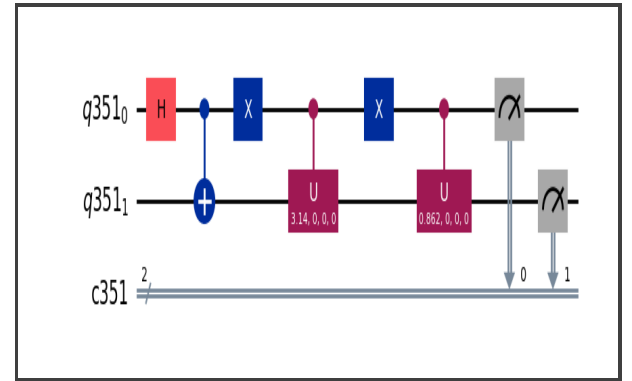


Circuit with appropriate U-gates

Figure 5.11: $t = 50$ units

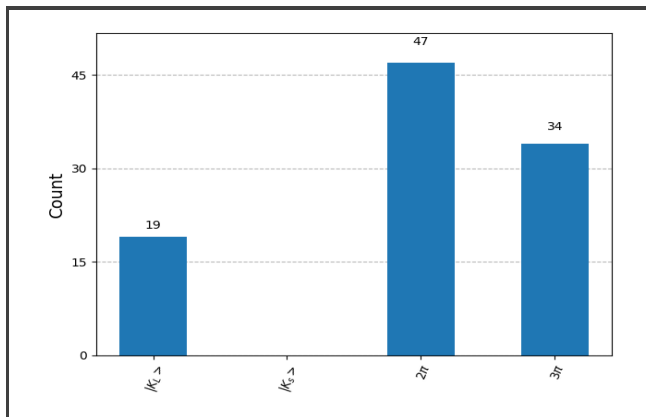


% of particles in each state

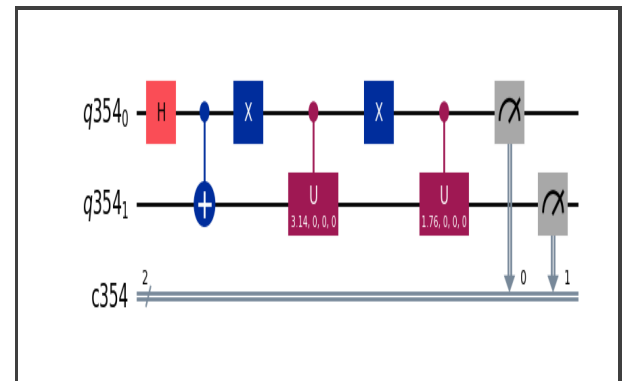


Circuit with appropriate U-gates

Figure 5.12: $t = 100$ units

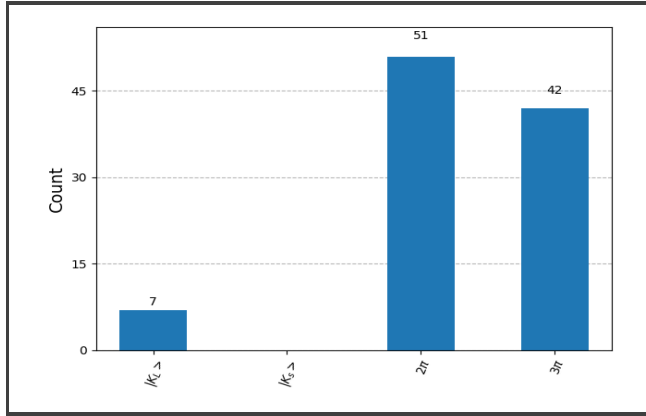


% of particles in each state

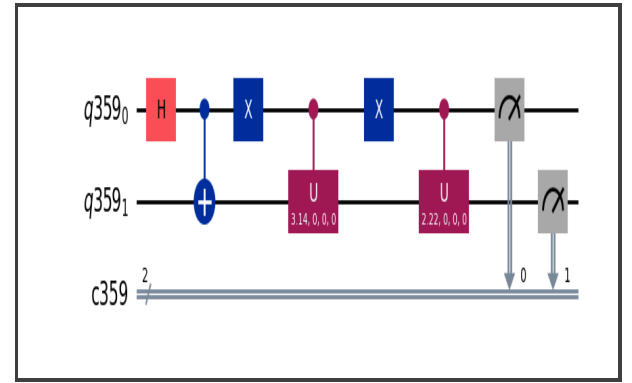


Circuit with appropriate U-gates

Figure 5.13: $t = 500$ units

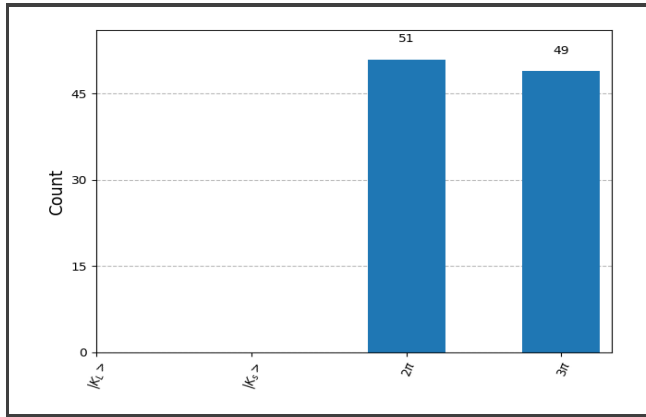


% of particles in each state

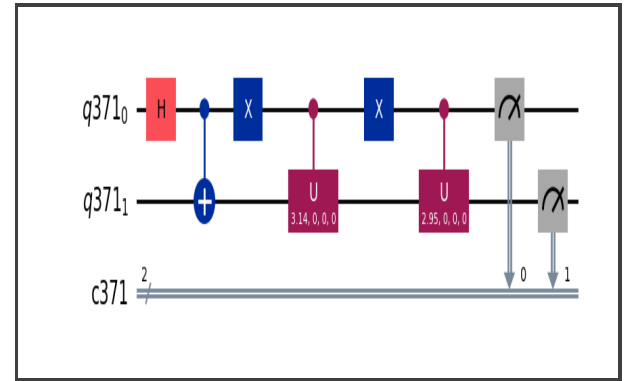


Circuit with appropriate U-gates

Figure 5.14: $t = 1000$ units

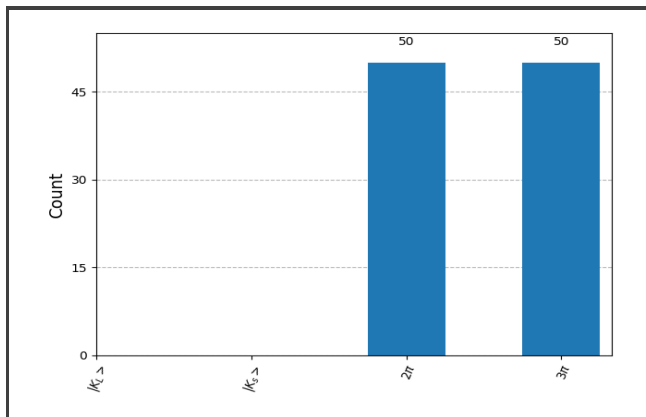


% of particles in each state

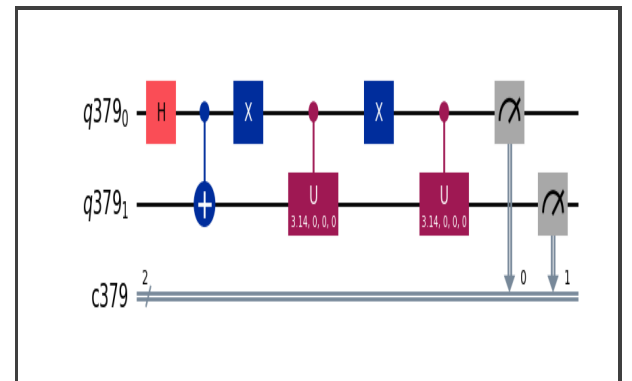


Circuit with appropriate U-gates

Figure 5.15: $t = 5000$ units



% of particles in each state



Circuit with appropriate U-gates

Figure 5.16: $t = 100000$ units

It is clearly seen with increasing time the angle of rotation approaches π and we get more decay into pions. The long component takes longer to decay as expected while the shorter component decays faster by a factor of 10^3 . ***Thus we have simulated Kaon decay through this quantum circuit.***

Chapter 6

Qiskit Simulation of Neutrino Oscillation

In this chapter we simulate Neutrino oscillation on a quantum circuit. We will simulate 2 flavour oscillations as studied in chapter 3.

6.1 Neutrino Oscillation of 2 Flavours

2 flavour neutrino oscillations involve a Hilbert space which is 2 dimensional. This can be represented on a single qubit, by the choice of basis as follows [11]:

$$|0\rangle = |\nu_e\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \text{and} \quad |1\rangle = |\nu_\mu\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Thus in this basis we can define the mass states as:

$$|\nu_1\rangle = U_{\text{PMNS}} |0\rangle \quad \text{and} \quad |\nu_2\rangle = U_{\text{PMNS}} |1\rangle$$

6.2 Defining the gates

Now we can easily define the PMNS matrix by the U -gate.

$$U_{\text{PMNS}} = U(2\theta, 0, 0) \quad \& \quad U_{\text{PMNS}}^\dagger = U(-2\theta, 0, 0) \quad (6.1)$$

The time evolution gate in this case is:

$$U(0, 0, \phi_i) = \begin{pmatrix} 1 & 0 \\ 0 & e^{-i\phi_i} \end{pmatrix}$$

As we need to consider only the relative phase we can apply the time evolution parameter ϕ_i as:

$$\phi = \frac{\Delta m^2 t}{2E}$$

As we had previously defined in 3.2.1 that we can interchange t for x in the time evolution of neutrino. Thus we can do an analysis for the spatial evolution of neutrino with this above gate.

6.3 Procedure

1. **Flavour state** Initiate with one of the flavour states, either $|0\rangle$ or $|1\rangle$. It is also possible to initialize with some arbitrary composition of both.
2. **Convert into the mass basis** By applying the PMNS matrix (U_{PMNS}) convert the flavour state into mass basis.
3. **Time evolution** Apply the time evolution gate to time evolve the mass basis.
4. **Revert to flavour basis** Apply $U_{\text{PMNS}}^\dagger$ to revert to the flavour basis.
5. **Measurement** Measure in the flavour basis to get the counts for the number of neutrinos of each flavour.

6.4 Quantum circuit

A typical quantum circuit in this case looks like:

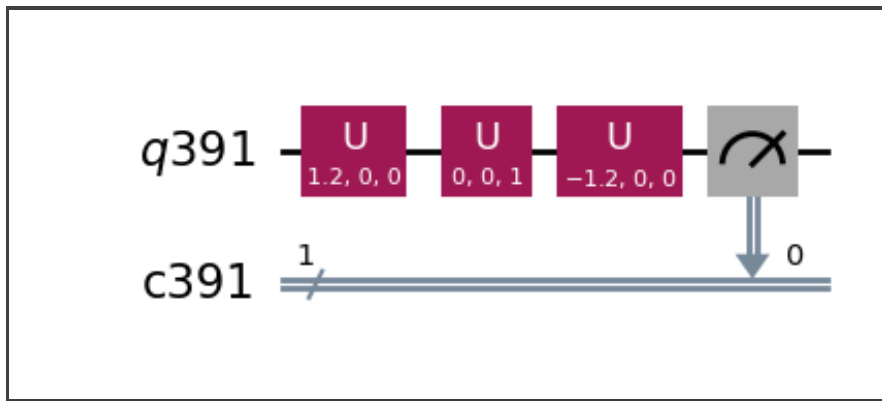


Figure 6.1: Circuit for a neutrino ν_e to ν_μ oscillation

The qubit is initiated as $|0\rangle$, that is ν_e . Here the first U gate from the left is the PMNS matrix. The next U gate is the time evolution gate with $\phi = 1$. The 3rd U gate is the $U_{\text{PMNS}}^\dagger$. And finally the qubit state is measured to a classical register.

6.5 Results

Each circuit is run for 100 shots and corresponding to each run a state of the qubit is measured. The count variable stores the state measured. As the total counts is 100, the results plotted in this section on the y axis gives the percentage. Simulating the same initial condition as depicted in fig 3.3, $\Delta m^2 = 0.0025 \text{ eV}^2$, mixing angle $\theta = 0.665 \text{ rad}$, $E_\nu = 1 \text{ GeV}$ we get:

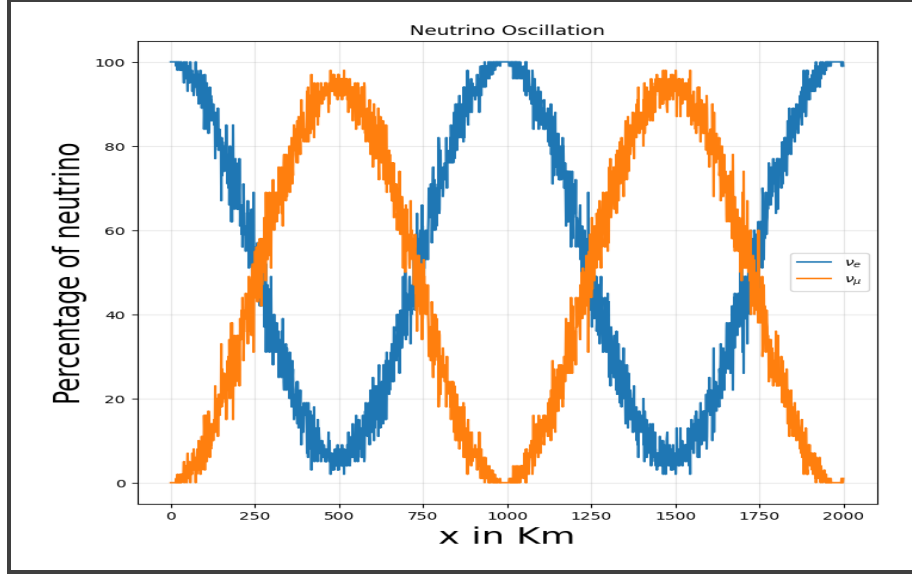
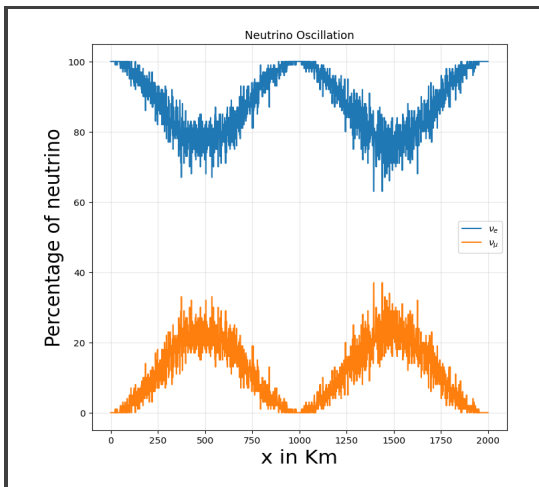
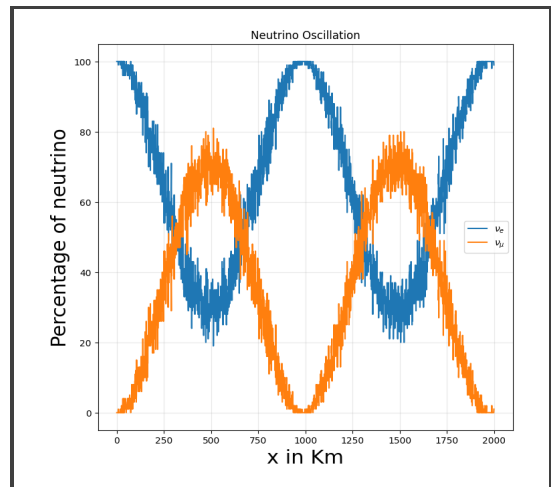


Figure 6.2: 2 flavour oscillation QISKIT simulation

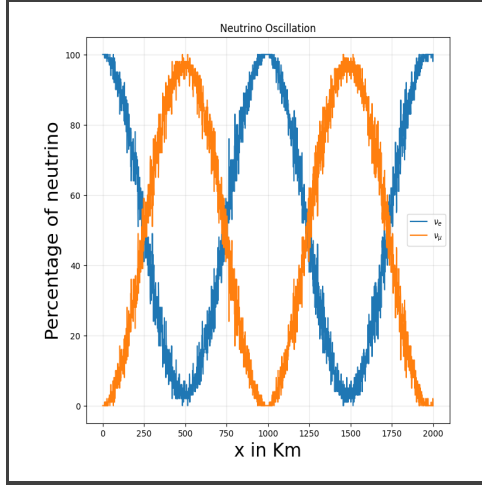
This verifies the nature of the plot and the equation 3.23. The error bars in the plot 6.2 are large because sampling has been done for all integral x (Km). By changing a particular parameter while keeping the rest fix we can study the oscillation of neutrinos. The following plots are given changing mixing angle while keeping the other parameters fixed:



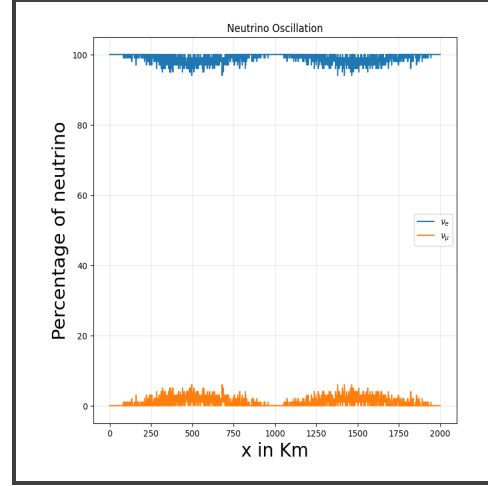
$$\theta = 0.25 \text{ rad}$$



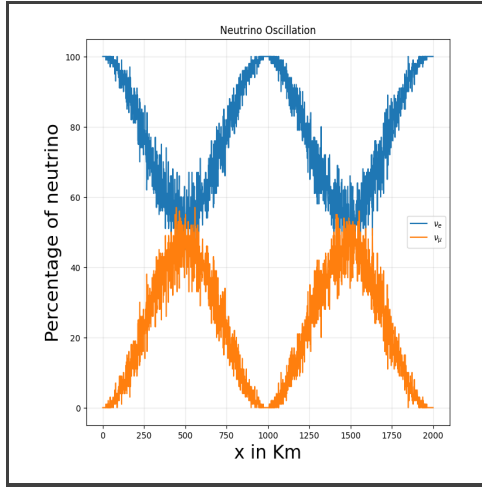
$$\theta = 0.5 \text{ rad}$$



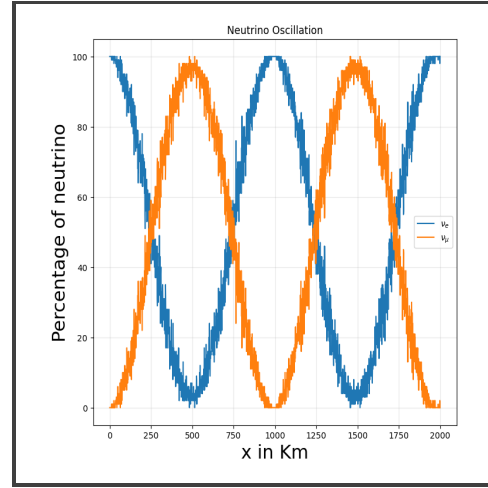
$$\theta = 0.875 \text{ rad}$$



$$\theta = 1.5 \text{ rad}$$

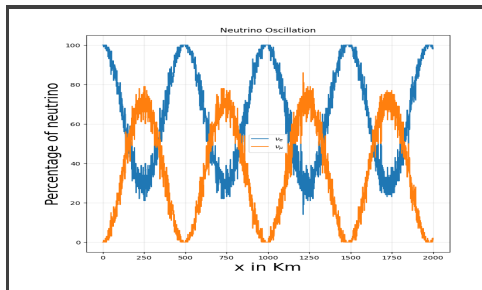


$$\theta = 1.94 \text{ rad}$$

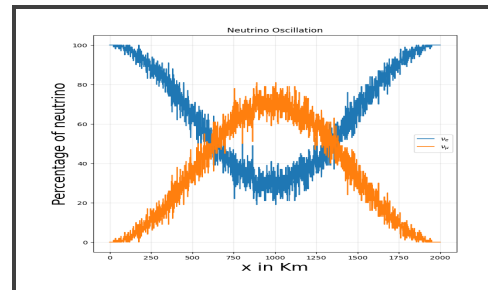


$$\theta = 2.5 \text{ rad}$$

From the plots it is clearly verified the $\sin^2(2\theta)$ relation of neutrino oscillation probability. Now lets take $\theta = 0.5 \text{ rad}$ and see the change in the oscillation probability with change in energy:



$$E = 0.5 \text{ GeV}$$



$$E = 2 \text{ GeV}$$

Figure 6.6: Energy dependence of Oscillation length

Clearly we can see the wavelength for oscillation depends on Energy.

6.6 Conclusion

Simulating in QISKIT helps us to map neutrino flavour oscillations to a Bloch-sphere rotation of appropriate basis vectors. Here we use the PMNS matrix to change between the mass and flavour basis. The time evolution gate applies the phase on the respective mass eigenstates. But as we are interested in the relative phase, in our simulation the relative phase is applied to one of the state. Then as we measure the state of the qubit back in the flavour basis we get composition of ν_e and ν_μ which is then plotted in terms of percentage. From the simulated results we can see that the plots acquired are indeed in accordance to the theory derived in chapter 3 but we still see large error bars due to the probabilistic nature of measurement in quantum computers.

Chapter 7

Summary

This report focused on the study of neutrino oscillations and how they differ from oscillations in the quark sector. The study began with a review of linear vector spaces, which forms the foundation of quantum mechanics. Then an example was worked out from classical mechanics to get familiarized with the matrix notation of vectors.

After establishing a mathematical framework the project proceeded to study the quark sector, specifically the neutral kaon system. Kaon oscillations were studied along with kaon decays. CP violation in this system was discussed in brief. The project then proceeded to the lepton sector. Here first 2 flavour Neutrino oscillations were discussed. Then 3 flavour neutrino oscillation was discussed where CP violation exists. After that a contrast was drawn between the two systems. The similarities and the differences were discussed.

Then the study focuses on simulating the oscillations on a quantum circuit. A brief introduction to quantum computing is provided. Qiskit module has been used to study and simulate kaon decays and Neutrino oscillations on a quantum computer. The results of the simulations have been plotted. This also brings us to the future aspects of the project. By changing other parameters in the simulation one can study the effect of mass difference in case of Neutrino oscillations. 3 flavour oscillations can also be studied using simulations on a quantum computer. CP violation can also be discussed in that context. We can also incorporate the CP violating parameters in the simulation of kaon decays. A combination of direct and indirect CP violating factors can be incorporated in the simulation but these were beyond the scope of this project. The future scope of this project would be to use a real quantum hardware to simulate the quantum circuits that have been simulated using IBM: Qiskit-Aer backend simulator service for this project.

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